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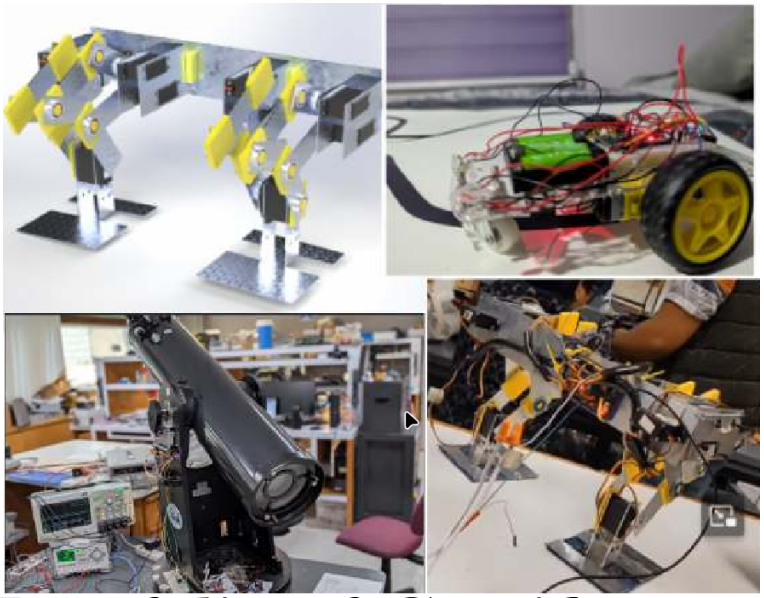

Portfolio of Certificates, Recognitions, and Projects

Ángel Hiram Alvídrez Hernández

2024-2026

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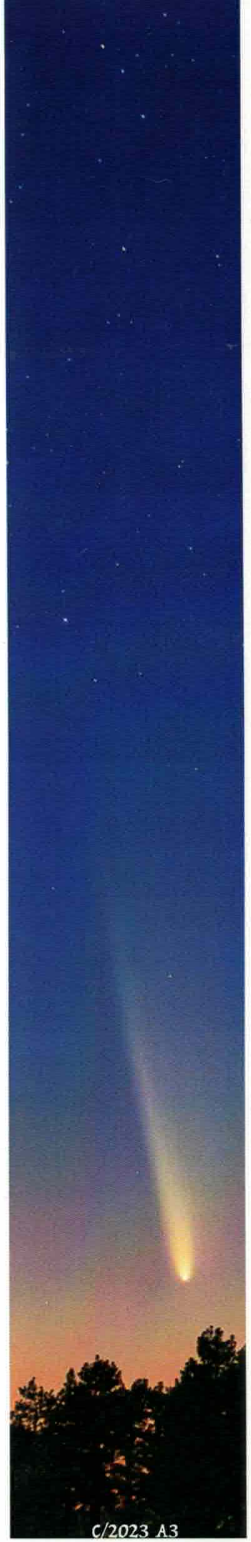

La Unidad Académica en Ensenada del Instituto de Astronomía y el Observatorio Astronómico Nacional San Pedro Mártir,
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junto al Comité Organizador del

XXXIV Verano Científico del
Observatorio Astronómico Nacional San Pedro Mártir

extienden la siguiente

Constancia

a

Ángel Hiram Alvídrez Hernández

Por su valiosa participación como **estudiante** en el Programa del XXXIV Verano Científico OAN-SPM, atendiendo cursos, talleres y conferencias, y desarrollando el proyecto de investigación que **le fue asignado** en el período comprendido entre el 16 de junio y el 4 de julio de 2025.

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IAD-990630-MK7-0013

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FUNDAMENTOS DE SEMICONDUCTORES

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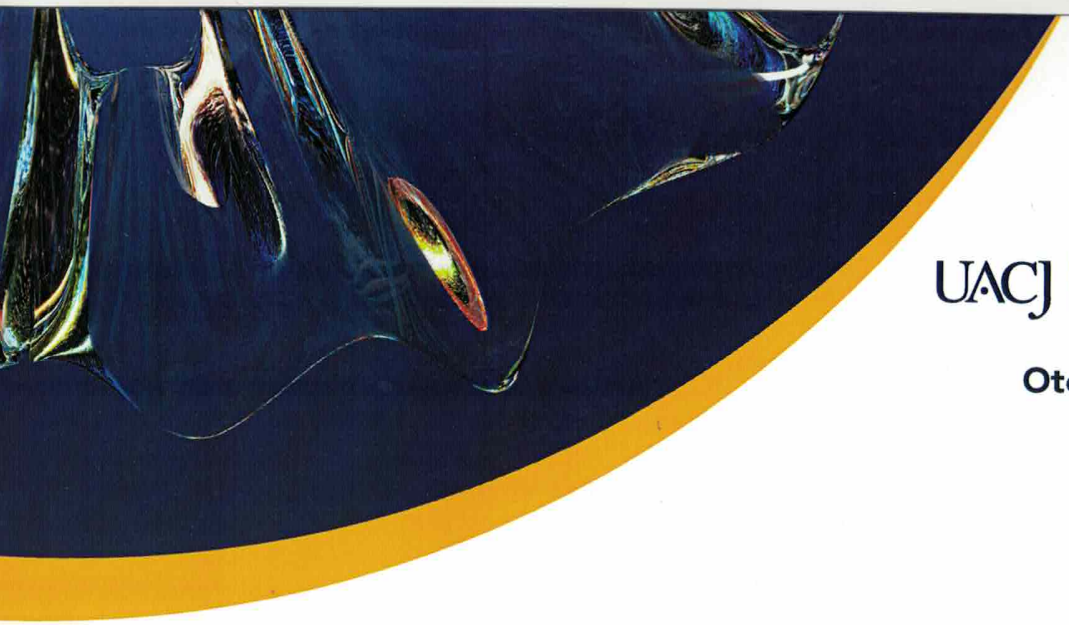

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Otorga el presente

RECONOCIMIENTO

A: **Ángel Hiram Alvídrez Hernández**

del programa

Ingeniería Mecatrónica

Por distinguirse como ESTUDIANTE INTEGRAL, con su excelente desempeño académico y por haber realizado actividades a favor de la comunidad durante el semestre enero-junio 2025.

Ciudad Juárez, Chihuahua, noviembre 2025.
Por una vida científica, por una ciencia vital

Dr. Daniel Alberto Constandse Cortez
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Dr. Salvador David Nava Martínez
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a través del Instituto de Ingeniería y Tecnología

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RECONOCIMIENTO

A

Ángel Hiram Alvídrez Hernández

Por su participación como ponente de la Conferencia:
"¿Estamos solos en el universo?",
durante la XXX Semana de Ingeniería.
Septiembre 2024

¨Por una vida científica,
Por una ciencia vital¨

Dr. Juan Francisco Hernández Paz
Director del Instituto de Ingeniería y Tecnología

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La Universidad Autónoma de Ciudad Juárez
a través del Instituto de Ingeniería y Tecnología

Otorga el presente

RECONOCIMIENTO

al Club de Astronomía

Por impartir la conferencia:
"Las muertes del universo".
Durante la XXIX Semana IIT

"Por una vida científica,
por una ciencia vital"

Dr. Juan Francisco Hernández Paz
Director del Instituto de Ingeniería y Tecnología

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Autonomous University of Ciudad Juárez

Robotics Laboratory

Department of Electrical and Computer Engineering

semester project

Motors and controllers
Edgar Alonso Martínez García

Team Members:

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Date: November 21, 2025

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1. Introduction

This project presents the development of an underactuated biped based on five-part mechanisms bars at each end, with a mechanical design modeled in CAD and an assembly perfected through successive stages of adjustment and improvement of the prototype aimed at achieving Controlled cyclical movements and initial walking tests. An Orange is also used. Pi as the main computer and an integrated IMU to record orientation and accelerations during static posture, cycloid trajectories and walking.

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2. 5-bar linkage

Two five-bar mechanisms arranged in parallel were implemented, one at each end.
This is an underactuated mechanism with two degrees of freedom. The active joints
They are actuated by a pair of MG996R servomotors, while the joints
Passive motion was achieved using ball bearings. Two more servomotors were added per limb, in
the connection to the hip and the connection to the robot's foot.

2.1. CAD Design

Figure 1 shows one of the five-bar mechanisms in isolation.
corresponding to the upper part of the limb. On the other hand, Figure 2 shows
the five bars with a color code, described in Table 1, along with the lengths of
each expressed in millimeters (mm). Once the CAD design of the mechanism is finished
With 5 bars, the design of the entire bipedal robot, shown in Figure 3, was completed.

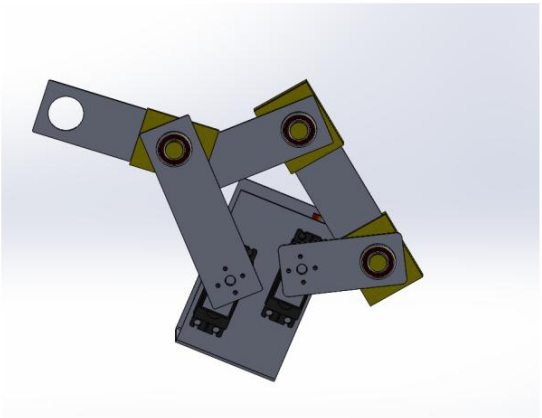

Figure 1: Design 1.

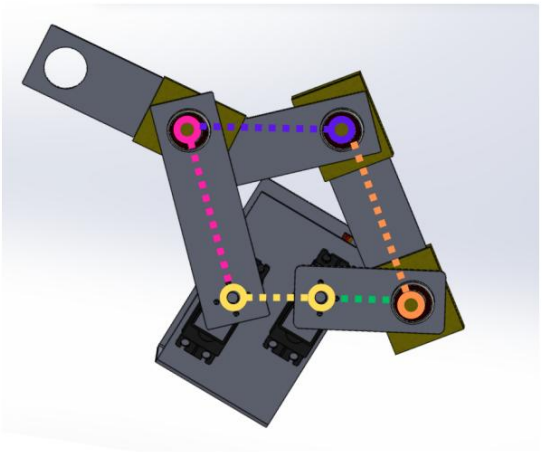

Figure 2: Definition of the 5 bars.

2.2. Kinematics of the mechanism

Once the five bars of the mechanism have been defined, the kinematic analysis is carried out with the
objective to determine the position, in millimeters, of the upper tip of the mechanism from
of the motor angles ÿ1 and ÿ4, as shown in Figure 4.

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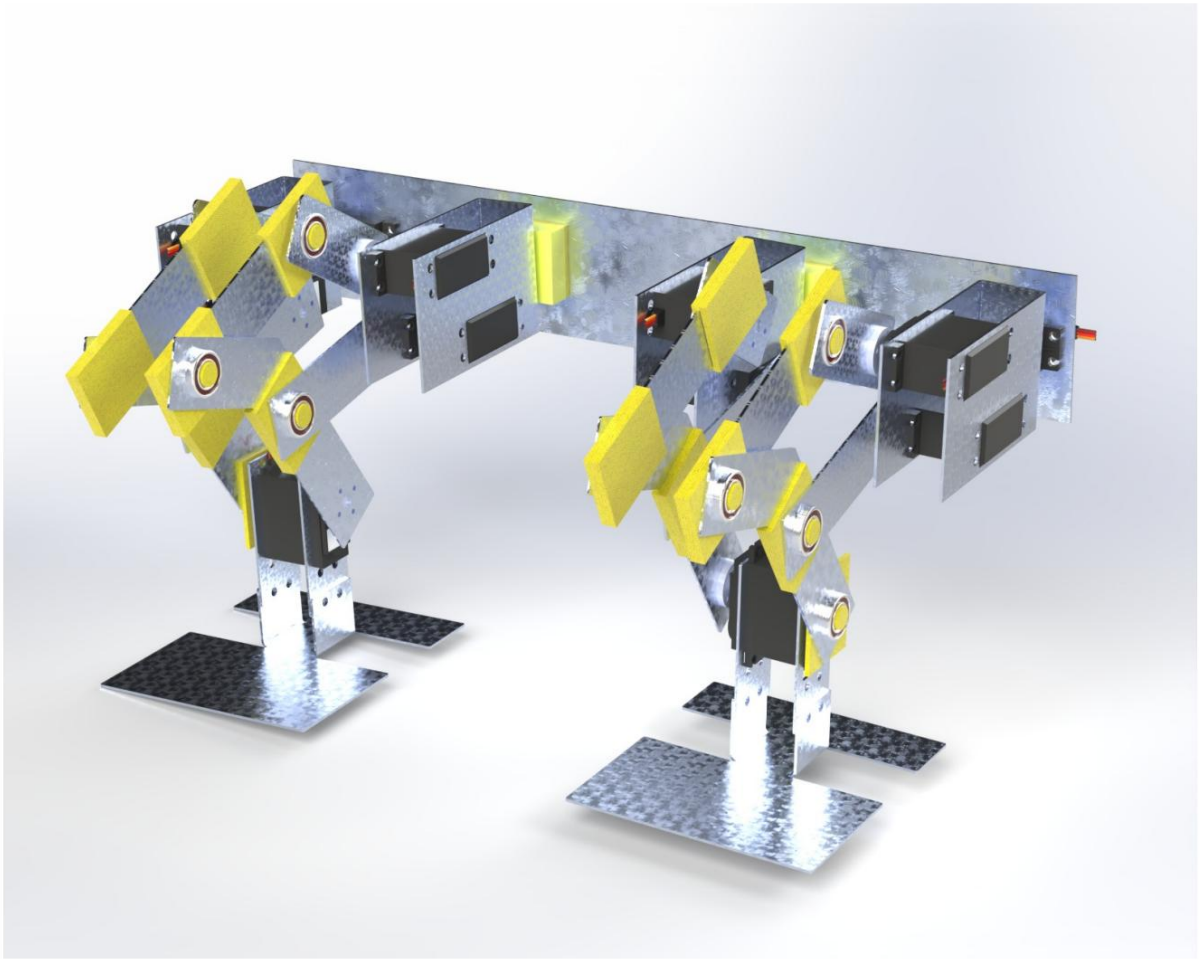

Figure 3: Design rendering.

Table 1: Table Title

Bar	Length	Color
L0	40.84 mm	Yellow
L1	43.5 mm	Green
L2	88.67 mm	Orange
L3	67.07 mm	Purple
L4	83 mm	Pink

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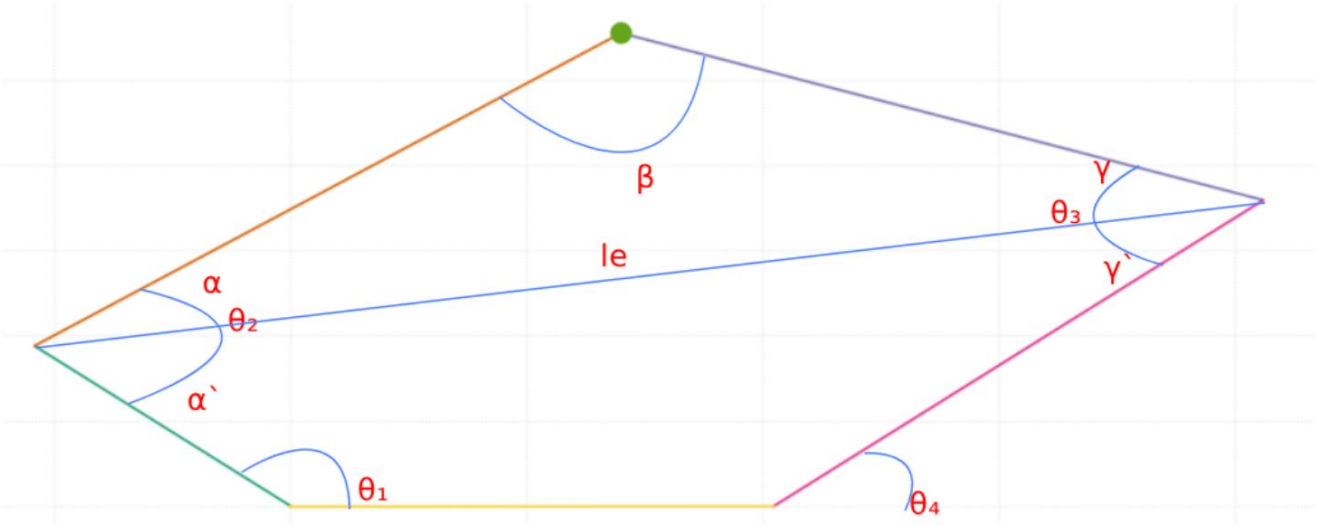

Figure 4: Visualization and nomenclature of the angles of the mechanism.

We begin by defining the position on the X and Y axes of the end of link L1 and L4 by the following expressions:

$$X1 = L1\cos(ÿ1);\ Y1 = L1\sin(ÿ1);$$

$$X4 = L4\cos(ÿ4);\ Y4 = L4\sin(ÿ4);$$

These positions are necessary to know the length of our "imaginary bar" Le, defined by:

$$Le = \sqrt{(X1 - X4)^2 + (Y1 - Y4)^2}$$

This allows us to see if the combination of angles ÿ1 and ÿ4 is possible without breaking our mechanism, because if the sum of L2 and L3 is less than our newly calculated Le, then The motors would be unable to reach the indicated angles without destroying the mechanism or themselves same.

Continuing with the analysis, we can use the law of sines and cosines and rearrange it to find

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the interior angles ÿ, ÿ and ÿ with the expressions:

$$ÿ = \arccos\left(\frac{Le_2 ÿL2\ _2 ÿL3\ _2}{ÿ2{\times}L2\ {\times}L3}\right)$$

$$ÿ = \arcsin\left(\frac{L3}{You}{\times}\sin(ÿ)\right)$$

$$ÿ = \arcsin\left(\frac{L2}{You}{\times}\sin(ÿ)\right)$$

To continue, we need to define another triangle by drawing an imaginary line, which
It goes from the top tip of L1 to the right tip of L0, as can be seen in the figure
5.

Figure 5: Visualization of L01 as the dotted line.

And using the law of cosines we can find the length of this imaginary line.

$$L01 = \sqrt{L0^{2\,2}{+}L1\quad ÿ2L1L0\cos(ÿ1)}$$

with which we can calculate the local version of the angle ÿ using:

$$ÿ' = \arccos\left(\frac{L01_2 ÿL4\ _2 ÿLe\ _2}{ÿ2LeL4}\right)$$

which helps us to find the internal angle ÿ3 by adding ÿ with ÿ ' and we will also need
the complement of these angles. With which we can finally arrive at the positions
Cartesian coordinates of the end effector of the 5-bar linkage with:

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$$PX = L0 + L4 \times \cos(ÿ4) + L3 \times \cos(ÿ4 + ÿ3 \quad ')$$

$$Py = L0 + L4 \times \sin(ÿ4) + L3 \times \sin(ÿ4 + ÿ3 \quad ')$$

Once we know a condition that limits our mechanism, we can calculate all the possible locations for our mechanism and to have our workspace secure, shown in figure 6

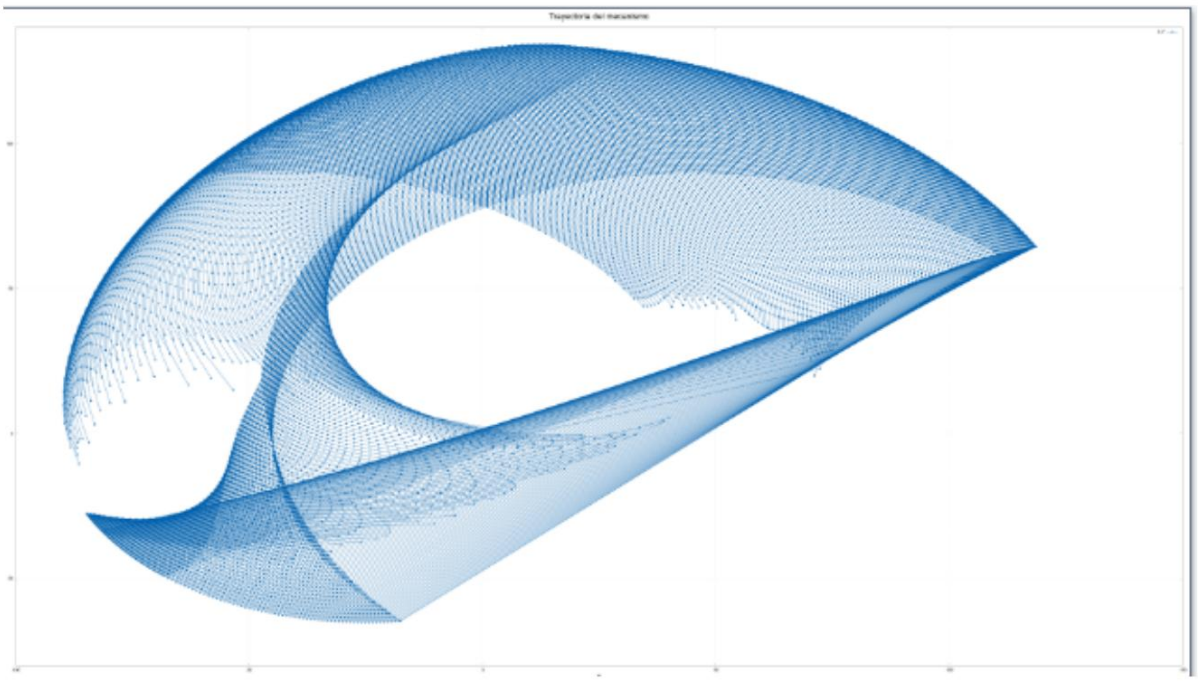

Figure 6: Workspace.

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3. Cycloid

The cycloid is the shape of motion that the five-bar linkage follows to simulate a hike.

In order for the mechanism to follow the desired movement, the following equations were applied to function of time (t):

$$ÿ1 = a+c×\sin(t) \quad (1)$$

$$ÿ4 = b+c×\sin t + 2 \frac{ÿ}{—} \quad (2)$$

where:

a : Initial angle of ÿ1,

b: Initial angle of ÿ4,

c: Oscillation amplitude.

3.1. Servo Movement

Once the joint movements were defined, the commands were sent to the servomotors. using an ESP-WROOM-32 and an expansion card with power supplies independent for control and power.

3.1.1. **Master** - **Slave** Communication

The computer embedded in the biped sends movement signals via serial port using the computer's USB ports and the control and sensing card.
A telegram is sent specifying which servomotor will be moved to what angle and with That's fast.

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3.1.2. Angular velocity

For controlling the angular velocity of the servos, the initial choice was to use

Libraries were used, but these did not provide good control of multiple servos simultaneously. To solve this

This led to the development of an algorithm;

$$\ddot{y}t = \ddot{y}0 + \sum_{k=1}^{N} \ddot{y}\,\mathrm{sgn}(\ddot{y}i+N\,\ddot{y}\ddot{y}i(k\,\ddot{y}1))\ddot{y}\ddot{y}i$$

Where ÿÿi are the angular intervals taken by the servo, calculated by:

$$\ddot{y}\ddot{y}i = \frac{1}{veli}$$

The time intervals were calculated using a microcontroller time counter,

instead of using interrupts, so that multiple servos can be controlled simultaneously.

All of this was done to create a constant and smooth speed in the servomotors, by sending them

to these angles by means of pulse width modulation, since it is the only data they can

receive the servos.[1]

3.2. Generated cycloid

Once we understand the kinematics of our mechanism and have control of the actuators

From this, the cycloid function was used and several iterations were performed in the

parameters of equations 1 and 2 until a response is obtained that allows the robot to move

maintaining its balance. The response of the end effector of the 5-bar linkage is shown

in figure 7

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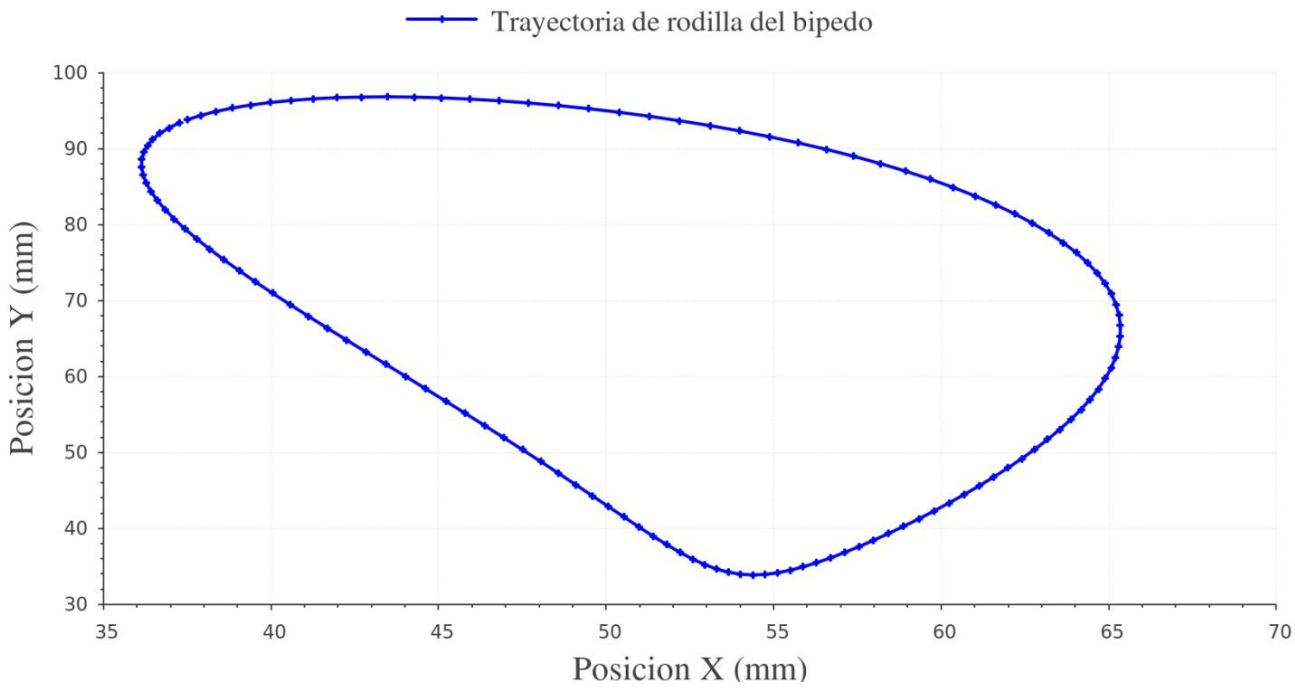

Figure 7: End effector of the 5-bar linkage.

4. Sensing

To prevent the Biped from falling, it needs to know that it is falling, so it is

I implement an IMU.

The selected IMU was the BNO80X, whose sensors are shown in Table 2

4.1. Data Fusion

The BNO08x is a System in Package (SiP) that includes a dedicated microcontroller and its

its own sensors. The secret to its performance lies in the fact that the microcontroller executes the

fusion algorithm within the chip, without the need for intervention from the main processor of the

user. [2]

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Table 2: IMU BNO08X

SENSOR	MEASURES	MAIN PROBLEM
Accelerometer	Gravity and linear acceleration.	It is sensitive to vibration noise.
Gyroscope	Angular velocity (rotation).	He/She suffers from drift.
Magnetometer: Measures the Earth's magnetic field (compass). It is very sensitive to interference from metals.		

4.1.1. Master - **Slave** Communication

Once our IMU is defined, it will communicate with the ESP-32 using the I2C protocol, and It will be constantly reading data from the IMU. Until the Orange Pi sends it a send a telegram requesting the IMU data, then the ESP-32 will send the latest data that has been stored in memory. This communication algorithm is visually exemplified in the Figure 8.

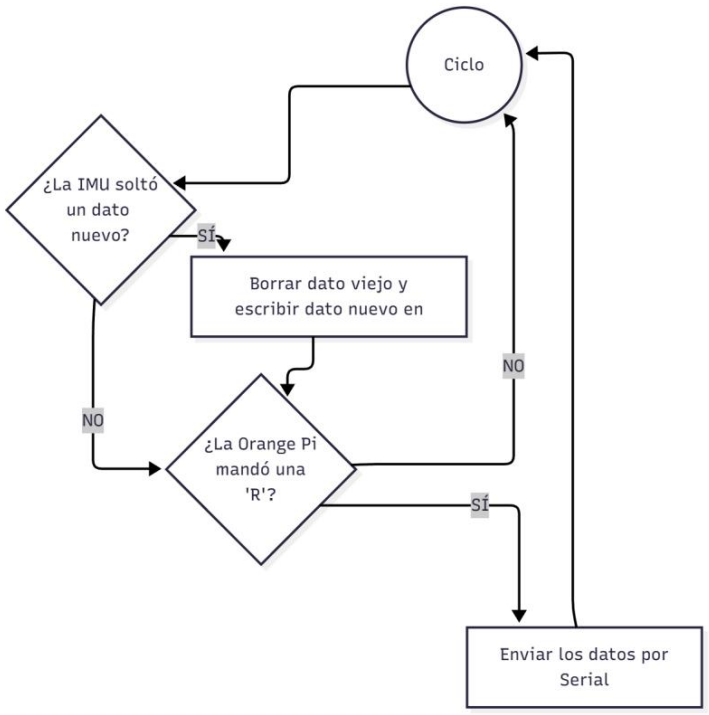

Figure 8: Slave ÿ Master communication sequence diagram.

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5. Control Strategy

The robot actively monitors its tilt in real time to make safety decisions, alternating between a Walking mode and a Recovery mode.

The control algorithm implemented in C++ executes the following logical loop in each iteration, also shown in Figure 9:

1. **Data Acquisition (Feedback):** The computer sends the data acquisition telegram data to the sensing module. The IMU subsystem responds by sending the angle of Current pitch.

2. **Stability Assessment:** The algorithm checks if the robot is inside a safe operating range, experimentally defined as ±20° of inclination in the Pitch axis.

3. **Decision Making:**

   - **Stable Case** (|Pitch| < 20°): If the pitch is safe, the system executes the next step of the cycloid trajectory. The coordinates (X,Y) are calculated, It solves the inverse kinematics and the servos move forward.

   - **Unstable Case (Fall Detected):** If the threshold is exceeded, the system is interrupted. temporarily stops the generation of the cycloid and activates a Safety Reflex:

     • If Pitch > 20° (Frontal Drop): The free limb moves rapidly forward to expand the support polygon.

     • If Pitch < ÿ20° (Backward Drop): The limb moves backward to counterbalance the fall.

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4. **Resumption of Movement:** The system maintains the safety position until the IMU readings return to the ±20° range . Once stabilized, the algorithm resumes the execution of the cycloid walk from the point where it was interrupted.

This approach allows for a predictable cyclical walk, delegating stability to the design. mechanical and passive filtering of disturbances.

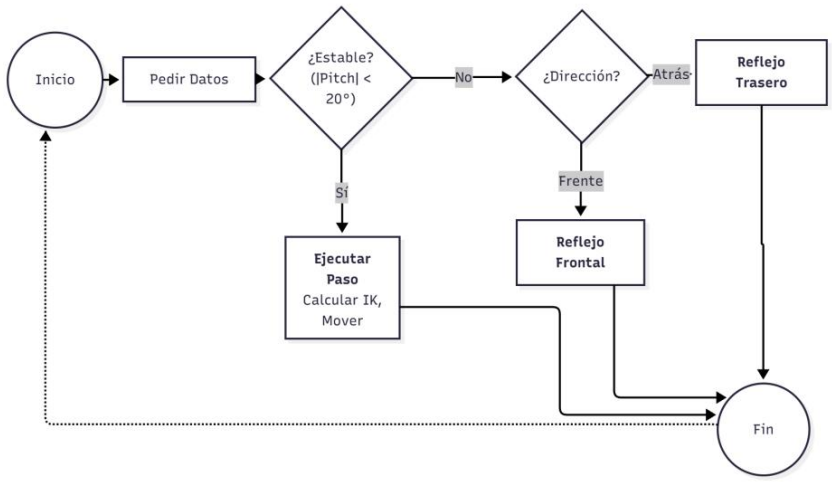

Figure 9: Walk sequence diagram

With this control architecture, even though it is somewhat "primitive", he managed to take a walk in the that the biped remained stable without needing support, thus managing to walk alone. [3].
While the walk was taking place, the IMU data was stored for later use.
graphed, which can be seen in figure 10

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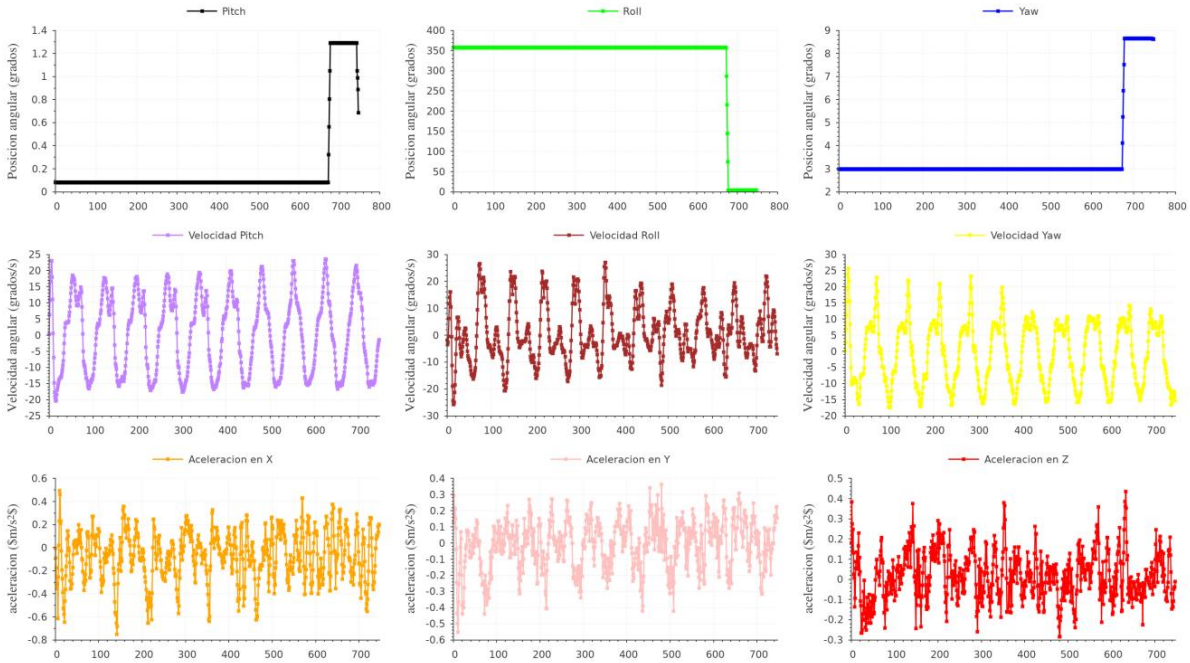

Figure 10: Walk sequence diagram

6. Conclusions

- Conclusion 1: The project to be carried out was the development of a bipedal walking robot, implementing a five-bar architecture for system kinematics, but
  We had problems with the structure of the biped due to its dimensions and weights.
  It was able to withstand it; the longest it took us to work on the biped was on the cycloid.
  that while we were looking for the right angles we had to be using trial and error to
  that he would walk the way we wanted, so that he would stay upright and
  could follow its path without tripping over its weight

- Conclusion 2: In summary, the project was a great experience working on making the biped since it requires solving both mechanical engineering problems (weight and structure) such as the precision in the implementation of mathematical functions (the cycloid) for to have a good walk.

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- Conclusion 3: Overall, the biped project achieved its main objective:
  generate a stable walk from a five-bar linkage controlled by
  a simple architecture, although there are still many improvements to be implemented.

  During development, several opportunities were identified that could optimize both the
  Performance, such as system efficiency, is a factor. For example, the capacity of
  neither the multiple serial ports available on the Orange Pi nor on the ESP32, which would have
  allowed to separate communication from sensing and control of servomotors, avoiding
  communication bottlenecks and potential blockages.

  Nor was the potential of the two cores of the ESP32 or the Orange Pi exploited, where
  One core could have been assigned exclusively to data acquisition and another to
  motion control, achieving truly parallel and smoother control.

  Another pending issue was the implementation of a Kalman filter, which would have allowed
  smooth the IMU data and generate a more accurate and controlled gait, instead of
  relying on a somewhat "brute force" strategy.

  Finally, I would have liked to perform a more in-depth dynamic analysis using the
  Lagrange's formalism, so that the biped could better take advantage of gravity and achieve a
  more natural and efficient movement.

  Even with these limitations, the project served as a solid foundation for understanding the
  integration between the mechanical, electronic and control parts of a bipedal robot, and leaves a
  a good starting point for more advanced versions in the future.

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Upgrading the control electronics of an 8-inch Orion telescope
Inches

This project involved modernizing the electronics of an Orion telescope whose original controller had failed due to environmental wear. This telescope has two encoders per axis: a high-resolution encoder for trajectories and fine object tracking, and a low-resolution encoder for pointing. Since the motors and encoders were still in good condition, they were reused, and a new control system based on a modular architecture was implemented.

Two ESP32 microcontrollers were used, one for each axis (altitude and azimuth). These microcontrollers control the movements of each motor with the help of an L298N H-bridge and process the optical encoder signals in real time. This required designing an electronic conditioning stage for each encoder, which includes signal amplification using an LM324 in non-inverting configuration, offset correction with a potentiometer, and digitization using inverter gates.

74HCT05.

The conditioning circuit was tested on a protoboard and then transferred to a printed circuit board that was made in the same laboratory.

Each ESP32 calculates trajectories based on encoder pulses and transmits position and status information to a Raspberry Pi via serial communication. This Raspberry Pi acts as the system's central node, running a Python script that receives coordinates and allows commands to be sent to the system.

In addition, Stellarium software is being integrated, which will allow users to select celestial objects in real time. Stellarium will send the coordinates directly to the Raspberry Pi. This feature is in its final stages of integration and testing.

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Audio amplifier based on transistors

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Abstract—This document presents the design, construction, and simulation of an audio amplifier based on transistors using a voltage divider configuration. The circuit incorporates an NPN 2N2222A transistor and a TIP32C transistor to amplify low-power audio signals intended for output devices such as speakers. The methodology includes detailed calculations and procedures to achieve an efficient design. Although the audio signals were amplified satisfactorily, the prototype reveals areas for improvement, such as thermal management, reduction of distortion, and precision in component selection, suggesting future optimizations for better performance and reliability. Among the topics addressed are transistor biasing, frequency response, and practical challenges in implementing electronic circuits.

Index Terms—Audio amplifier, transistors, divider of voltage, electronics, transistor biasing, frequency, speakers, capacitors, resistors.

I. INTRODUCTION

A transistor-based audio amplifier is an electronic circuit designed specifically to increase the power of the audio signals it receives, ensuring playback on output devices or speakers, allowing the audio to have higher volume and clarity. As the name suggests, these devices are based on the principle of amplification, where a transistor controls a larger current according to a much weaker input signal [1], [2]. This same principle often has different applications in areas of technology such as sound systems, musical instrumentation, cell phone sound, among others.

One of the main characteristics of an amplifier is its great ability to process low-power signals and convert them into stronger signals that can be controlled via speakers. This is achieved by using transistors, allowing multiple configurations, in which highlight the common-emitter, common-collector and push-pull configurations. Each of these configurations has specific characteristics that make them suitable for different amplification stages [3], [4].

A. Principle of operation of a transistor

The transistor falls into the category of being a device semiconductor that has three terminals; emitter, base and collector [1]. In Fig. 1 the visual diagram of bipolar transistors is observed.

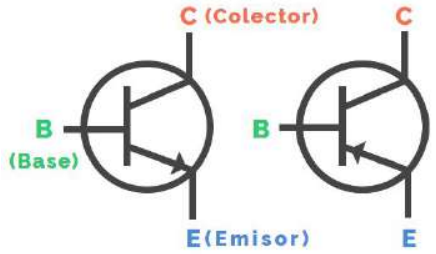

Fig. 1: Diagram of bipolar transistor symbols: NPN (left) and PNP (right), showing their terminals: Base (B), Collector (C) and Emitter (E) [5].

The main function of a transistor is to control the flow of current, acting as a switch or, alternatively, as an amplifier. When amplifying a signal, the transistor is able to increase the amplitude of a small signal applied to its base, resulting in a higher power signal at the collector [1].

The relation between a transistor and an amplifier is that the transistor has its three terminals that can be designed as amplifiers with specific characteristics of gain, impedance, and frequency [3].

While a transistor is an individual component capable of controlling and amplifying signals, an amplifier is a complete circuit that works with transistors, among other components, in order to increase the amplitude of an input signal according to the desired needs [4].

II. THEORETICAL FRAMEWORK: AUDIO AMPLIFIER BY MEANS OF A VOLTAGE DIVIDER

There are different configurations when amplifying a signal; in this section we discuss an audio amplifier based on a voltage divider, which basically is a circuit that uses this specific configuration to establish the necessary bias level in the transistor, allowing it to operate efficiently in the active region. This configuration is characterized by being simple and reliable, since the voltage divider, formed by two resistors in series, divides the supply source to generate a constant voltage that controls the transistor's behavior [2].

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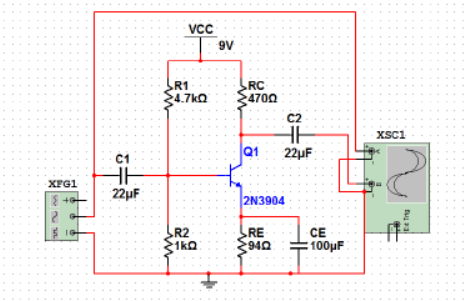


Fig. 2: Schematic of an audio amplifier based on a voltage divider using a 2N3904 transistor. Includes resistors for biasing (R1 and R2), collector resistor (RC), emitter resistor (RE), a decoupling capacitor (CE) and coupling capacitors (C1 and C2) for input and output signal, powered by a 9V source [6].

Figure 2 shows the schematic of an audio amplifier based on a voltage divider using an NPN transistor model 2N3904. The circuit uses resistors for biasing, coupling and decoupling components, and a 9V power supply.

A voltage divider is typically composed of two resistors in series connected between the supply voltage (VCC) and ground. The voltage at the junction of these resistors is used to bias the transistor's base. If R1 and R2 are the divider resistors, the bias voltage is given by:

$$VB = \frac{R2}{R1 + R2} VCC \quad (1)$$

This voltage determines the base current (IB) of the transistor, which in turn controls the collector current (IC) according to the relation:

$$IC = \beta IB \quad (2)$$

where β is the current gain of the transistor.

A. Components

The circuit also includes additional resistors and capacitors to improve its performance:

- RE and RC: The RE resistor in the emitter helps stabilize the operating point, while RC determines the amplifier's gain.
- CE: The bypass capacitor CE removes the alternating current component in RE, increasing the amplifier's gain.
- C1 and C2: The coupling capacitors allow only alternating current signals to pass between the input and output stages, blocking any direct current component.

The frequency response of the amplifier is determined by the capacitors and resistors in the circuit. The lower cutoff frequency (fc) of a capacitor C with a in series resistor R is calculated as:

$$fc = \frac{1}{2\pi RC}. \quad (3)$$

This parameter is crucial to ensure that the amplifier operates in the frequency range (20 Hz - 20 kHz).

III. METHODOLOGY

To carry out this project, a structured but practical process was followed [7], which covers from the initial idea to the validation of the final design. Below, the stages carried out are described.

A. Circuit design

For the circuit design of an audio amplifier based on transistors, a biasing network using a voltage divider served as the basis. To achieve more efficient results in the project, two amplifiers were built, meaning the circuit design was divided into two phases.

1) First stage of amplification: In the first stage of amplification, the circuit sketch was made (Fig. 3.) and afterwards the relevant calculations. In this phase, the guitar input is present, followed by the connection of a transistor in voltage division with capacitors.

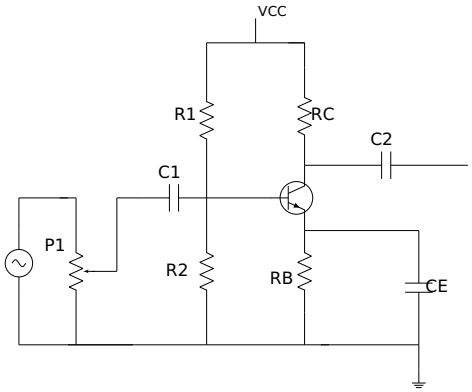


Fig. 3: Initial schematic of the first amplification stage. This preliminary circuit defines the basic configuration for signal amplification, including the use of a transistor as the active element and passive components such as resistors and capacitors to stabilize and control the gain. The values of the components, such as R1, R2, RE, C1, C2, and CE, have not yet been determined, allowing for later adjustment to meet design specifications.

In the first amplification stage, a 2N2222A transistor is used as a preamplifier, which is expected to achieve a collector current gain of 100. For this, suitable resistors were chosen to ensure a collector current of 15 mA and a base current of 150 µA. To select these resistors, the analysis began by setting the collector and emitter pins at 50% of the supply voltage to allow a signal range with minimal distortion, in this case 18V, and the emitter resistor is expected to have 10% of the supply voltage. Therefore, for

determine the value of the required resistance, the following calculation was carried out.

$$RE = \frac{10\% \times 18V}{15mA} = 120\Omega \quad (4)$$

For the collector resistor it was assumed that the current in the collector and emitter is the same (IE = IC); to obtain 50% of the supply voltage across the emitter-collector, the collector resistor must drop 40% of the supply voltage, making a calculation similar to the previous one, we obtain:

$$RC = \frac{40\% \times 18V}{15mA} = 480\Omega \quad (5)$$

For the resistors of the voltage divider located at the base of the transistor we ensured that R2is 10 times larger than the emitter one, guaranteeing a base current smaller than that in the collector and emitter.

$$R2 = 10 \times Re = 1.2K\Omega \quad (6)$$

And for the last resistor, the analysis of the voltage that will drop on the base of the transistor was performed; this voltage is known by Kirchhoff's laws, since the voltage at the emitter pin of the transistor was chosen to be 10% of our voltage source, and the base-emitter voltage drop is approximately 0.7V; with this voltage we can calculate the expected voltage drop on the first resistor and thus calculate it with the voltage relation between base, the expected drop, and R2.

$$Vbe \approx 0.7V, Ve = 1.8V \quad (7)$$

$$Vb = 1.8 + 0.7 = 2.5V \quad (8)$$

$$VR1 = 18 - 2.5 = 15.5V \quad (9)$$

$$R1 = \frac{15.5}{2.5} \times R2 = 10.33K\Omega \quad (10)$$

Finally we calculate the power each resistor will dissipate to ensure we buy the right type.

$$PRE = 1.8V \times 15mA = 27mW \quad (11)$$

$$PRC = 7.2V \times 15mA = 108mW \quad (12)$$

$$PR2 = \frac{2.5V2}{1200\Omega} = 5.2mW \quad (13)$$

$$PR1 = \frac{15.5V}{10.333K\Omega} = 23.25mW \quad (14)$$

2) Second amplification stage: For the amplifier of power, we chose the TIP32-C transistor because it has a higher current capability and a moderate power dissipation, but it can be improved up to 20 times with a heat sink; this transistor was given a collector current of IC = 150mA and its base current was adjusted to IB = 15mA to be connected to the first amplification stage.

The procedure used to determine the component values in the first amplification stage was applied, analogously, to calculate the values corresponding to the components in the second amplification stage. This allowed ensuring continuity in the design and coherence in the circuit performance.

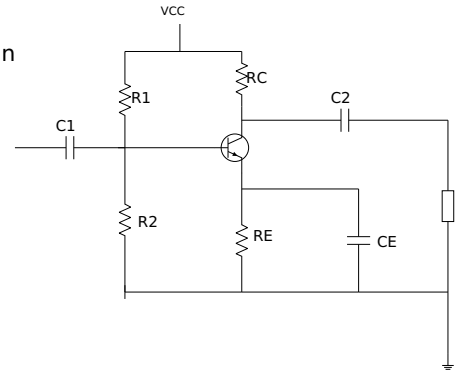


Fig. 4: Schematic of the final amplification circuit. This design shows the complete configuration of the second amplification stage

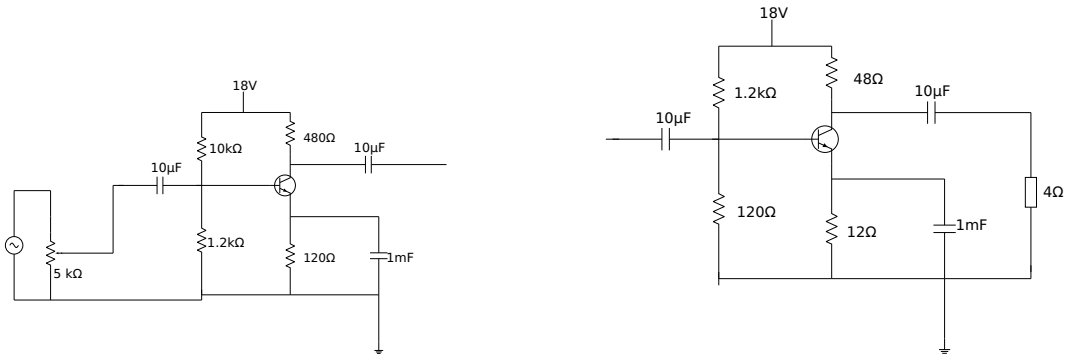
amplifier, where R1 and R2 form a voltage divider to bias the transistor. Capacitors C1 and C2actuate as coupling for the input and output signal, respectively- while CE provides a higher gain by decoupling the RE resistor. The circuit is designed to provide a stable and efficient amplification.

3) Final circuit design: The final circuit design (Fig. 5) was the result of a tuning process based on the previous amplification stages. The main objective was to optimize the total system gain and ensure the circuit was stable and efficient. This design incorporates carefully selected components to meet the specific amplification needs.

B. Circuit Simulation

The circuit simulation in its final stage was performed using the free Tinkercad software, a very useful tool for designing and simulating electronic circuits in a visual and interactive way. The implemented design includes a 2N2222A transistor and a TIP32C, which fulfill the fundamental role for audio amplification, chosen because their gain characteristics and power handling capacity (Fig. 6.).

The circuit configuration includes the voltage divider mentioned earlier, formed by resistors at the base of the 2N2222A transistor, as this sets the proper bias point for the transistor to operate



(a) Circuit 1 (b) Circuit 2

Fig. 5: Amplification circuit in two stages. (a) First stage of amplification: the initial circuit design is presented, where the component values, such as resistors and capacitors, were previously calculated to ensure an efficient increase of the initial signal. This stage establishes the basic gain of the system. (b) Second stage of amplification: shows the final circuit design, where the calculated component values optimize the total gain and stability of the system. This stage reinforces the amplification achieved in the first, ensuring an output signal suitable for the desired purpose.

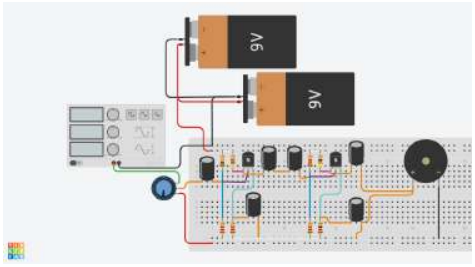


Fig. 6: Visualization of the simulation of the final stage of the amplifier circuit, made in Tinkercad.

Fig. 7: The transistor in the image has the same package (TO220) as the TIP32C used in the circuit design [8].

appropriately. On the other hand, coupling capacitors (C1 and C2) were used to ensure a clean transfer of the alternating current signal between each layer, thus removing the undesired direct current from other components. Additionally, a decoupling capacitor (CE) was integrated, which helps to eliminate the voltage drop in the emitter resistor, thus maximizing gain.

For the power supply, it was determined to use 18V, so two 9V batteries were chosen, which provide the energy needed for the circuit to operate.

C. Construction of the circuit

After simulating the circuit and determining the optimal values for each component, it was assembled on the breadboard, subsequently performing the tests and adjustments necessary to ensure proper operation.

IV. RESULTS

With the construction of the audio amplifier, practical tests were carried out to evaluate its behavior under real conditions. The circuit was connected to a speaker

Radox 065-176 of 5 inches with a 15W rating and an impedance of 4Ω, reproducing audio signals from an electric guitar. The results showed that it successfully amplifies the audio signal, allowing its output at a perceptible volume. However, distortion is noticeable at the output. Additionally, an increase in the temperature of the transistors and some resistors was observed during prolonged operation. Even with the use of a heat sink, it is not enough to keep the temperature steady.

in a range optimal, evidencing the need for improvements in the thermal management of the circuit. This high thermal dissipation affects the efficiency and durability of the amplifier, requiring the use of more powerful heat sinks. In general, the amplifier meets its function, but it presents aspects that require improvement in output quality and performance during prolonged use without heating up as much.

V. ANALYSIS OF RESULTS

The built audio amplifier showed acceptable performance, capable of amplifying audio signals from an electric guitar. However, the results analysis led us to identify areas where improvements are needed for much more optimal and reliable performance.

For example, although the amplification was sufficient to reproduce a good sound volume, harmonic distortion was detected. Also, during prolonged use, the TIP32C transistor showed heating, along with the closest resistors even with a heat sink, and this could significantly reduce the useful life of the component and the proper functioning of the amplifier. Or the robustness of the design, which although allowed us to validate the functionality of the circuit, the protoboard is not so practical; ideally, it would be a small and compact design on a PCB, which would have much more stable connections and avoid noise or false contacts. In addition, during the construction stage while selecting the components, it was not possible to obtain

resistors of 48Ω exact ones, so two connected in resistors of 100Ω parallel were used to obtain an approximate value to the 48Ω. Similarly, for 12Ω resistors it was necessary to use some of 24Ω in parallel. These are modifications that slightly affect the behavior of the circuit and should be improved. In short, although the objectives were met, improving all these aspects will make the amplifier more efficient

and stable, suitable for more demanding uses.

VI. CONCLUSION

The design, simulation, and construction of the audio amplifier based on transistors managed to fulfill its main objective of amplifying audio signals for playback at an adequate volume in output devices, such as speakers. The implementation of the amplification stages, using the 2N2222A and TIP32C transistors, allowed for correct gain and satisfactory performance. However, during the analysis of results, it was determined which areas need to be improved for better amplifier performance. During testing, heating of the TIP32C transistor was observed, even when integrating a heat sink, which compromises the thermal efficiency of the circuit. Likewise, the quality of the output signal showed slight distortions. These issues are significant for modifying the design for better thermal resistance. Another important aspect was the selection of components, specifically the resistors. Not having used the exact resistors that were calculated due to complications related to availability in stores led to the use of parallel combinations, which has slight repercussions on the design's precision. Finally, although the use of the protoboard is practical for checking the amplifier's functionality, for more reliable and durable use, it was determined that ideally it should be transferred to a PCB to make it more compact and with more precise connections to avoid false contacts. In conclusion, the development of this audio amplifier based on transistors enabled the exploration of both the fundamental principles of amplification and the practical challenges that arise in the implementation of electronics. Although the project achieved satisfactory basic functionality, it showed several technical aspects that require

improvements to increase the system's performance.

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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 14 Controlling the position of a cart

Sensors and Instrumentation

Angel Alvidrez 225581:

Date: April 30, 2025

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1. Objective

Develop a position control system for a motorized cart, using a
an ultrasonic sensor and a PID controller implemented on Arduino, in order to
maintain a fixed distance from a wall.

2. Introduction

In the field of mechatronics and automatic control, one of the fundamental challenges
It is about ensuring that a system maintains a desired position or trajectory in the face of potential
External disturbances. In this exercise, a position control system is implemented.
for a cart with direct current motors, whose objective is to maintain a distance
constant with respect to a wall using an HC-SR04 ultrasonic sensor.
The system is controlled by a PID (Proportional, Integral, Derivative) algorithm.
which dynamically adjusts the speed and direction of the cart to minimize error
between the current and desired distance. The latter is established remotely through
via a Bluetooth interface, allowing real-time experimentation with different values
reference.

The use of PID control in positioning systems is widely used in industry
Due to its simplicity and effectiveness, this practice allows the student to understand
to practically explain the operation of a closed-loop controller, as well as the
importance of properly adjusting its parameters to obtain a stable response and
precise.

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3. Definitions and conceptual framework

3.1. Ultrasonic sensor

This sensor allows measuring distances through the time of flight of an ultrasonic pulse.

By sending a high-frequency sound pulse and detecting the echo reflected off an object, it is

It is possible to calculate the distance using the formula:

$$\text{distance} = 2 \quad \frac{340\ \text{m/s} \cdot t(\mu s)}{} \cdot \frac{1\ \text{s}}{1{,}000{,}000\ \mu s} \cdot \frac{100\ \text{cm}}{1\ \text{m}}$$

Which, simplified, becomes

$$\text{distance (cm)} = t(\mu s) \times 0.01715$$

Figure 1: HC-SR04 Ultrasonic Sensor

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3.2. PID Control

The PID (Proportional, Integral, Derivative) controller is one of the most used in system control. Its output is based on three components:

- Proportional (P): responds proportionally to the current error.
- Integral (I): accumulates the error over time to correct persistent errors.
- Derivative (D): anticipates the error trend to reduce oscillations.

The combination of these three actions allows for a balance between speed and stability. and accuracy. In this case, the PID is implemented with the ARDUINO PID v1 library in the ESP 32, configuring the parameters empirically until a behavior is achieved desired.

3.3. Bluetooth Communication

Bluetooth communication allows data to be sent and received wirelessly. In this In practice, the ESP32 integrated module is used along with the BluetoothSerial library. to receive the setpoint value in real time from an external device, allowing modify the position reference without reprogramming the system using a Android application created in MIT App Inventor.

3.4. Direct Current (DC) Motors

Direct current motors convert electrical energy into rotary motion. They are widely used due to their ease of control and low cost. In this practice, They used 6V motors, which allow for acceptable speeds for a laboratory or classroom environment.

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Figure 2: 6V DC Motor

3.5. PWM Signals

Pulse width modulation (PWM) is a technique that allows the speed of a DC motor to be controlled by adjusting the duty cycle.
cycle) of a digital signal. The higher the duty cycle, the greater the energy supplied to the engine, and therefore, its speed.

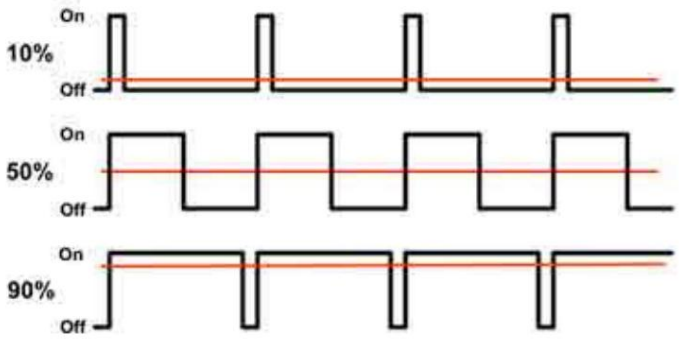

Figure 3: PWM Graph

3.6. Bridge H (L298N)

The H-bridge is a circuit that allows changing the polarity of the power supply to the motors, which allows their direction of rotation to be reversed. The L298N is a dual-bridge motor that

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It allows you to control two motors independently, and can also control their speed using the PWM signal sent from a microcontroller such as the Arduino.

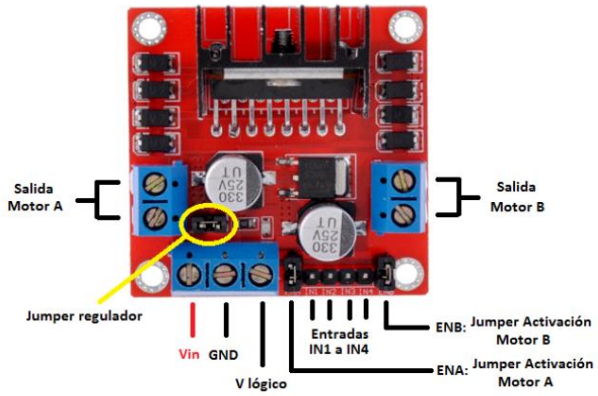

Figure 4: H-bridge L298N

4. Development

The position control system was implemented on an ESP32 board, which was responsible for both from the reading of the HC-SR04 ultrasonic sensor and from the control of the motors through the L298N H-bridge. The programming was done in the Arduino environment for Controlling an ESP32, using the PID_v1.h library to implement the controller PID and the BluetoothSerial.h library to receive data from a mobile device or computer via Bluetooth.

The program flow is as follows:

- Initialization of serial and Bluetooth communication.
- Pin configuration for motor control and ultrasonic sensor.
- Waits for a setpoint value sent via Bluetooth, which represents the distance desired distance in centimeters between the cart and the wall.
- Constant distance reading using the time of flight of the ultrasonic sensor.
- Calculation of the control signal using the PID algorithm.

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- Application of the control signal to the motors, using PWM to adjust the speed according to the magnitude of the error.
- If the calculated PWM is very low (but not zero), a minimum value is imposed for to ensure that the engines respond.

The code was adjusted empirically, mainly the values of the constants of the PID and the minimum PWM threshold needed for the cart to move, because with values At very low speeds, the cart did not move. It was observed that, with the constants Kp = 1.0 and Ki = 0.0 and Kd = 0.2, the system's behavior was sufficiently stable and responded appropriate form for position corrections.

5. Code

This was the code used, as explained above:

Listing 1: PID control code for the cart

```
# include < PID_v1.h >
#include          "BluetoothSerial .h"
BluetoothSerial SerialBT;

// L298N motor pins
const int M1_A = 16                ,    M1_B = 17;
const int M2_A = 18                ,    M2_B = 19;

// HC - SR04 Pins
const int trigPin = 27                    ,    echoPin = 26;
long duration;
double readDistance() {
    digitalWrite(trigPin,                 LOW ) ;
    delayMicroseconds (2) ; // Optional, but can help stabilize
    digitalWrite(trigPin,                 HIGH);
    delayMicroseconds (10) ;
    digitalWrite(trigPin,                 LOW ) ;
    long duration = pulseIn ( echoPin                    ,    HIGH); // no timeout
    return duration * 0.01715;
```

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```
}

// PID
double Setpoint , Input , Output ;
double Kp = 1.0 , Ki = 0.0 , Kd = 0.2;
PID myPID (& Input , & Output, & Setpoint, Kp, Ki , Kd , DIRECT);

// Minimum motor threshold
const int motorMinPWM = 135;

void setup() {
  Serial . begin (115200) ;
  while (! Serial ) delay (10) ;
  Serial.println ("\n--- Distance PID Control ---") ;

  SerialBT . begin (" ESP32_Carrito ") ; // Bluetooth Name

  // Motor pins
  pinMode ( M1_A , OUTPUT ) ; pinMode ( M1_B , OUTPUT ) ;
  pinMode ( M2_A , OUTPUT ) ; pinMode ( M2_B , OUTPUT ) ;
  // Ultrasonic pins
  pinMode (trigPin , OUTPUT ); pinMode ( echoPin , INPUT ) ;

  // Wait for first Bluetooth Setpoint
  Serial. println ("Waiting for Bluetooth Setpoint (1 byte in cm)... ");
  while ( SerialBT . available () == 0) delay (10) ;
  Setpoint = SerialBT . read () ;
  Serial. print ("Initial setpoint received: ") ;
  Serial . print ( Setpoint ) ;
  Serial . println (" cm") ;

  // Init PID
  Input = readDistance() ;
  myPID . SetMode(AUTOMATIC);
  myPID . SetOutputLimits ( -255 , 255) ;

  Serial. println ("Setup complete. Starting control... ");
```

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```
}

void loop () {
    // Update Setpoint if a new one arrives
    if ( SerialBT . available () > 0) {
        Setpoint = SerialBT . read () ;
        Serial.print ("New Setpoint received: ");
        Serial . print ( Setpoint ) ;
        Serial . println (" cm") ;
    }

    // 1) Read distance
    Input = readDistance() ;
    Serial. print ( "Current Dist: ") ; Serial. print (Input); Serial. println ("cm" )

    // 2) Calculate PID
    myPID . Compute() ;

    // 3) Motor control
    int pwm = abs (( int ) Output ) ;
    bool forward = (Output > 0) ;

    if ( pwm > 0 && pwm < motorMinPWM ) pwm = motorMinPWM ; // Apply minimum PWM

    if (ahead) {
        analogWrite ( M1_A , pwm ) ; analogWrite ( M2_A , pwm ) ;
        analogWrite ( M1_B , 0) ; analogWrite ( M2_B , 0) ;
    } else {
        analogWrite ( M1_B , pwm ) ; analogWrite ( M2_B , pwm ) ;
        analogWrite ( M1_A , 0) ; analogWrite ( M2_A , 0) ;
    }

    // 4) Debug in serial monitor
    Serial. print ("SP:" ) Serial. print ( Setpoint ) ;
    Serial. print ("PV:") ; Serial. print (Input);
    Serial. print ("Err:") ; Serial. print ( Setpoint - Input ) ;
    Serial. print ("Out:" ) Serial. print ( Output ) ;
```

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```
    Serial. print ("PWM:" ) Serial. println ( pwm ) ;

    delay (100) ;
}
```

6. Results

During system testing, a range of desired distances between 10 cm and
50 cm, dynamically transmitted via Bluetooth. Within this range, the cart maintained a
stable position facing the wall, continuously correcting its distance by means of the
PID controller.

It was observed that, when the desired distance was increased beyond 50 or 60 cm, the sensor
The ultrasonic scale was beginning to show measurement errors. This is probably due to
The sensor operates with a detection angle of approximately 30°, so
At large distances, it may begin to detect the floor instead of the wall, affecting the
reading accuracy.

Within the optimal range (10–50 cm), the system's behavior was smooth, without oscillations.
very marked or very abrupt movements. It was also verified that the use of a threshold
A minimum PWM (motorMinPWM) was required to prevent the motors from stalling.
static with very small signals, especially when the error was low but still
It needed correction.

The PID output signal adjusted well to the setpoint changes, allowing
the cart adapts to new positions with a quick and stable transition. All the
The system's behavior could be monitored in real time from the serial monitor.

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7. Conclusion

The implemented system successfully controlled the position of a cart relative to
to a wall using an ultrasonic sensor and a PID controller. The integration of the
The Bluetooth module allowed dynamic modification of the setpoint, which facilitated experimentation.
with different target distances without needing to modify the code or restart the system.
The ultrasonic sensor was found to be reliable within an approximate range of 10 to 50 cm.
but it begins to fail at greater distances due to its conical detection shape,
which can cause erroneous readings by picking up echoes from the floor or other objects. This highlights
the importance of knowing the physical limitations of the sensors when designing
a control system.

The use of the PID controller allowed for smoother behavior of the cart, adjusting its
position continuously without needing to stop completely. Empirical adjustments to
The constants Kp, Ki, and Kd were key to achieving stable and responsive performance.
Finally, this exercise was useful for reinforcing concepts of automatic control and electronics.
and applied programming, showing how to integrate multiple components into a system
functional.

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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 16 Line-following cart

Sensors and Instrumentation

Angel Alvidrez 225581:

Date:April 30, 2025

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1. Objective

Develop a functional prototype of a line-following cart using sensors
infrared and DC motors controlled by PWM, using a
ESP32 microcontroller. The system must be able to detect a trajectory in a
track and follow it autonomously, demonstrating the principle of feedback between
sensors and actuators.

2. Introduction

A line follower is a type of mobile robot that uses optical sensors to detect
and follow a defined path, usually a line of contrasting color (black on
(white background). These systems are one of the most common applications in robotics
educational due to its ability to demonstrate fundamental concepts of electronics,
control and programming in a practical way.
In this task, a prototype of a line-following cart was developed using a microcontroller.
ESP32, infrared sensors consisting of a photodiode-phototransistor pair, and motors of
Direct current controlled by pulse-width modulation (PWM). The system
It was designed to follow a track, reacting to changes in ground reflectivity.
to adjust the vehicle's movement in real time.

3. Definitions and conceptual framework

3.1. Photodiode

The FD-IR333C is an infrared light emitting diode that typically operates at a voltage direct current of approximately 1.2V. Its function is to emit infrared radiation which will reflected by the surface under the sensor. Light surfaces reflect more light, while that dark cells absorb a large part of the radiation. In this project, the photodiode is connects in series with a 100 ohm current-limiting resistorΩ,forming part of sensor pair that detects the line on the track.

Figure 1: Transparent Photo Diode (FD-IR333C)

3.2. Phototransistor

The PT331C is a phototransistor that responds to infrared radiation emitted by the photodiode. When reflected infrared light strikes its base, the phototransistor allows the flow of current from the collector to the emitter. In this configuration, the collector is connected to a 10k pull-up resistorΩtowards 5V and also to the analog input of the ESP32, while the emitter is connected to ground. This arrangement allows for the detection of changes in the amount of reflected light as variations in output voltage, which the microcontroller interprets to determine the position of the cart with respect to the line.

Figure 2: Phototransistor (PT331C)

3.3. Direct Current (DC) Motors

Direct current motors convert electrical energy into rotary motion.

They are widely used due to their ease of control and low cost. In this practice,

They used 6V motors, which allow for acceptable speeds for a

laboratory or classroom environment.

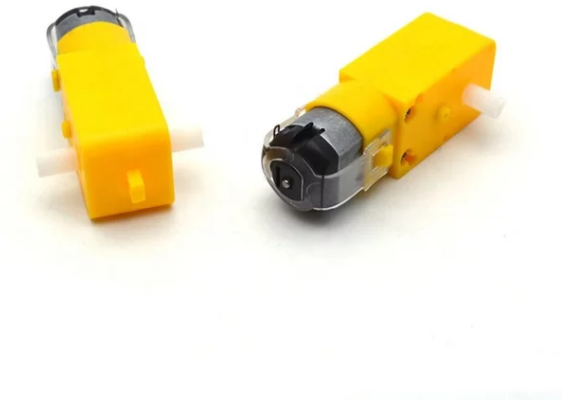

Figure 3: 6V DC Motor

3.4. PWM Signals

Pulse width modulation (PWM) is a technique that

It allows you to control the speed of a DC motor by adjusting the duty cycle.

cycle) of a digital signal. The higher the duty cycle, the greater the energy supplied to the engine, and therefore, its speed.

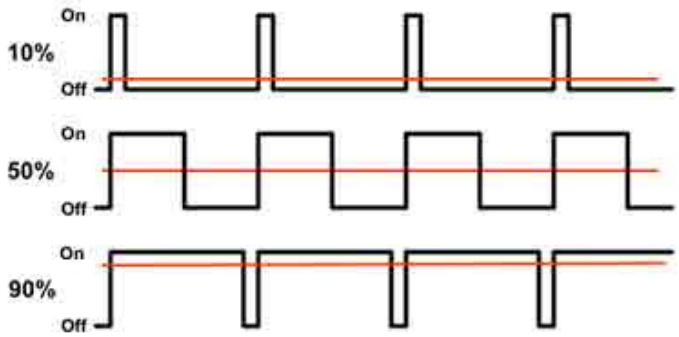

Figure 4: PWM Graph

3.5. Bridge H (L298N)

The H-bridge is a circuit that allows changing the polarity of the power supply to the motors, which allows their direction of rotation to be reversed. The L298N is a dual-bridge motor that It allows you to control two motors independently, and can also control their speed using the PWM signal sent from a microcontroller such as the Arduino.

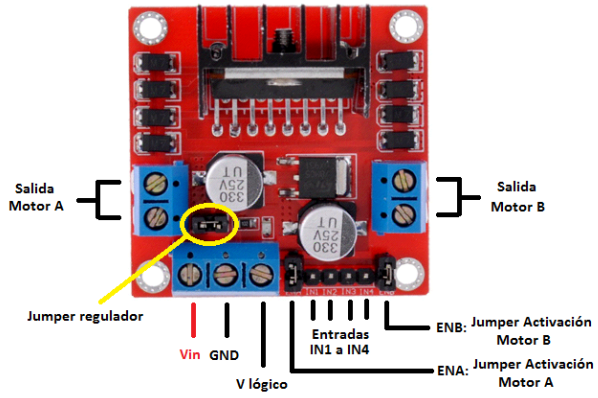

Figure 5: H-bridge L298N

4. Development

For line detection, two sensor pairs were used: each consisting of a FD-IR333C photodiode and a PT331C phototransistor.

For the photodiodes, a 100 ohm limiting resistor is usedΩwith a diet of 5V and an approximate voltage drop of 1.2V, with both diodes in parallel which divides the current for each LED in half. Applying Ohm's Law:

$V_{led} = 5V - 1,2V = 3,8$

$I_{led} = \frac{V_{led}}{R2} x_1 = \frac{3,8V}{100\Omega} x_1 \approx 20mA$

This configuration ensures stable operation and adequate current consumption to protect both the LEDs and the ESP32.

The phototransistors were connected with the collector to a 100k pull-up resistorΩ which goes to 5V, and also to an analog input of the ESP32 (GPIO34 and GPIO35).

The emitter of each phototransistor is connected directly to ground. Thus, when the sensor It detects a light surface (which reflects more IR), the phototransistor conducts and the voltage across The input pin decreases. When it detects a black line (less reflection), conduction decreases and the input voltage rises, allowing the position of to be clearly distinguished the line.

The system has two direct current motors, one for each wheel. These motors They were connected directly to the GPIO16, GPIO17, GPIO18 and GPIO19 pins of the ESP32 through a driver that allows H-Bridge switching for each motor.

Each motor receives two signals: one PWM control signal for speed, and another logic signal for the direction of rotation. The left motor is controlled by pins 18 and 19, and the right one by pins 18 and 19. through pins 16 and 17.

The main algorithm consists of constantly reading the analog values of the two sensors and compare their values against a threshold (approximately 1900). If the sensor The left side detects a line and the right side does not, the cart turns left; if the opposite occurs, Turn right. If both sensors detect a line or bottom, the cart stops.

Motor speed is controlled by PWM, with a value of 255 (maximum speed).

5. Code

This was the code used, as explained above:

Listing 1: Line follower cart code

```
#include<Arduino.h>

// Sensors        analgesics Left
const   int       sensor          =  35;      //   GPIO34   (ADC1_CH6)
const   int    sensorDer          =  34;      //   GPIO35   (ADC1_CH7)

// Engine A (left wheel) const
        int    motorA_IN1         =  18;      //   GPIO18
const   int    motorA_IN2         =  19;      //   GPIO19

// Engine B (right wheel) const
        int    motorB_IN1         =  16;      //   GPIO16
const   int    motorB_IN2         =  17;      //   GPIO17

// ======== PARAMETERS ======== const
        int    THRESHOLD          =  1900;
const   int    SPEED              =  255;

voidsetup() {
    Serial.begin (115200);
    pinMode(sensorIzq ,         INPUT);
    pinMode(sensorDer ,         INPUT);
    pinMode(motorA_IN1 ,          OUTPUT);
    pinMode(motorA_IN2 ,          OUTPUT);
    pinMode(motorB_IN1 ,          OUTPUT);
    pinMode(motorB_IN2 ,          OUTPUT);
}

voidloop() {
    intleftval = analogRead(leftsensor);
    intvalDer = analogRead(sensorDer) -300;
```

```
    Serial.printf("Left Sensor = %4d                Sensor Der= %4d\n", valIzq , valDer);

if(leftval < THRESHOLD && rightval > THRESHOLD) {
        leftturn();
    }
    else if(leftval > THRESHOLD && rightval < THRESHOLD) {
        turn();
    }
    else{
        stall ();
    }
delay (1);
}

voidleftturn() {
    analogWrite(motorA_IN1 ,        0);             digitalWrite(motorA_IN2 ,          LOW);
    analogWrite(motorB_IN1 ,        SPEED);         digitalWrite(motorB_IN2 ,          LOW);
}

voidturn() {
    analogWrite(motorA_IN1 ,        SPEED);         digitalWrite(motorA_IN2,           LOW);
    analogWrite(motorB_IN1 ,        0);             digitalWrite(motorB_IN2,           LOW);
}

voidstall () {
    digitalWrite(motorA_IN1 ,         LOW);   digitalWrite(motorA_IN2 ,          LOW);
    digitalWrite(motorB_IN1 ,         LOW);   digitalWrite(motorB_IN2 ,          LOW);
}
}
```

6. Results

Upon completion of the implementation of the line-following cart, its functionality was verified.
on a track with a black line on a white background. The cart was able to identify
correctly the line and react according to the programmed logic. When the sensor
When the left side detected the line, the cart turned left; when it was the right side,
It turned to the right; and when both sensors were out of line, the cart
It stopped. The response was quick and stable during testing.
The infrared sensors responded with clear and distinguishable readings: high values
when there was no line (white background) and low values when detecting the black line. This
It allowed a reliable detection threshold to be established in 1900, with minimal calibration.
in addition to the right sensor (-300) to compensate for differences in sensitivity. Furthermore,
The PWM control provided a constant and suitable speed for the test without causing
no sudden movements or deviations off the track.
During testing, the cart completed multiple routes without losing its path or needing
External intervention. The system proved efficient for basic monitoring tasks.
in line, fulfilling the objectives established for the practice.

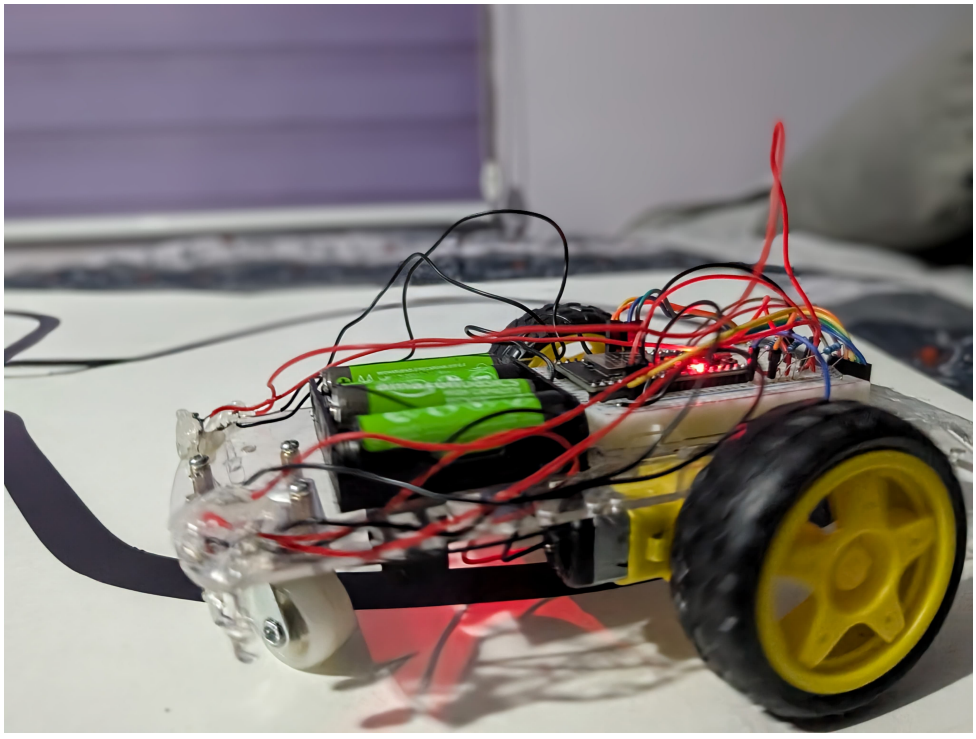

Figure 6: Cart in operation

7. Conclusion

The line-following cart project met the established objectives in a
satisfactory. The combination of infrared sensors (FD-IR333C and PT331C), the microcontroller
The ESP32 and direct current motors made it possible to create an autonomous navigation system
capable of following a line with precision.
The use of pulse width modulation (PWM) facilitated efficient control of the
Motor speed, which improved the cart's stability. Analog reading
The infrared sensors allowed for the reliable detection of variations between background
white and the black line, which made it possible for the cart to react correctly to
the different situations that arose on the track.
The system showed consistent behavior in the tests, performing precise turns.
and stopping when necessary. Although the system is functional, problems can arise.
improvements, such as the implementation of adaptive cruise control or the incorporation
of additional sensors to improve accuracy on tighter curves.
Overall, the project was successful in its implementation and represents a solid foundation.
for future robotics and automation projects.

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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 15 Turning a light bulb on and off with applause

Sensors and Instrumentation

Angel Alvidrez 225581:

Date: May 6, 2025

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1. Introduction

In this practical exercise, a system capable of turning on and off is designed and implemented. A 110V spotlight is activated by applause detection. This is achieved using a chain signal processing that ranges from sound capture to switching
Using a triac with an Arduino Uno, the circuit is expected to:

- Detect the sound of applause and convert it into a processable electrical signal.
- Filter and amplify only the frequency band where applause predominates (3 kHz - 6 kHz).
- Generate an adjustable digital pulse for the Arduino, based on the intensity of the applause.
- Count the number of claps in a 3-second interval to decide when to turn on. (2 claps) or off (4 claps) of the spotlight.
- Switch a triac (2N6071) with an MOC3022 optocoupler to isolate the signal from 110 VAC line control.

The gain is adjusted with a potentiometer in the feedback network (Rf), allowing to control the amplitude of the signal coming from the KY-038 microphone. The circuit with +5V from the Arduino and -5V generated with an L7905 regulator connected to a 9V battery.
The filter was implemented with a second TL081 designed to attenuate all frequencies outside the range of interest, ensuring that only applause generates a significant signal.
A third TL081 functions as a simple window comparator. The trigger threshold is Adjust with another potentiometer so that the output saturates and delivers a digital level (HIGH) when the filtered signal exceeds the reference level.
The comparator output is connected to pin 2 of the Arduino, which detects rising edges. to increase an applause counter.
A 3-second interval is measured (starting with the first clap) within which: 2 claps the Arduino turns on the simulation LED (and sends the signal to the MOC3022 to turn on the triac).

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Four claps turn the Arduino off (and stop triggering the triac). Values exceeding
Applause or the end of the 3 seconds restarts the count. The Arduino will control an MOC3022.
which opto-isolates the control signal and triggers the 2N6071 triac, responsible for switching the light
110 VAC.

2. Definitions and conceptual framework

2.1. KY-038 Microphone

It is a sound sensor module with analog and digital output. In this project, only the sound sensor module is used.
It uses the analog output, which provides a signal proportional to the intensity of the
sound captured. The sensor consists of an electret condenser microphone and a small
integrated amplifier. The signal is weak and requires additional amplification to be useful.
in signal processing.

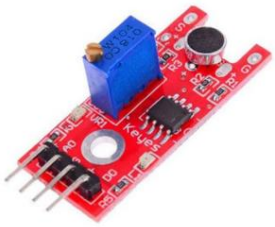

Figure 1: HC-SR04 Ultrasonic Sensor

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2.2. TL081 operational amplifier

It is a low-noise operational amplifier with differential input and single output.

He used this component for all analog stages of the circuit:

- Initial amplification of the microphone signal
- Bandpass filter
- Level comparator

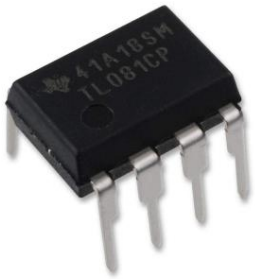

Figure 2: Op Amp TL081

2.3. Active bandpass filter

Applause has a strong energy component between 3kHz and 6kHz. To be sure

that only these frequencies pass through and the system does not react to ambient noise or

For other voices, an active bandpass filter with op amps was designed.

This filter attenuates signals outside that band, dissipating low noise (speech, ambient).

and high (sparking, interference).

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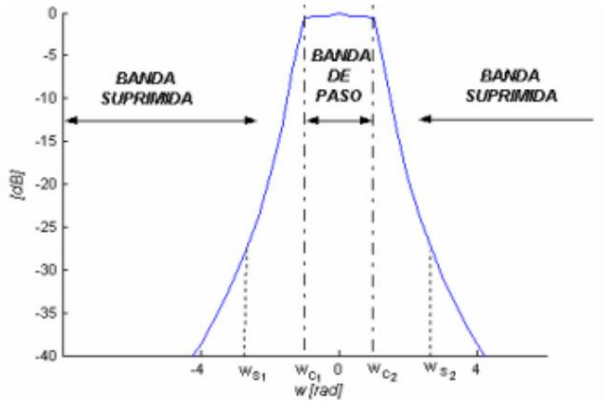

Figure 3: Output of a bandpass filter

2.4. Comparator with adjustable reference level

The filtered signal passes to a comparator built with another TL081, which is compared against an adjustable reference voltage with a potentiometer, when the signal exceeds that When the threshold is reached (by applause), the op amp saturates and its output abruptly changes to a high level (5V), acting as an applause detector.

The output is a digital signal that the Arduino can easily read.

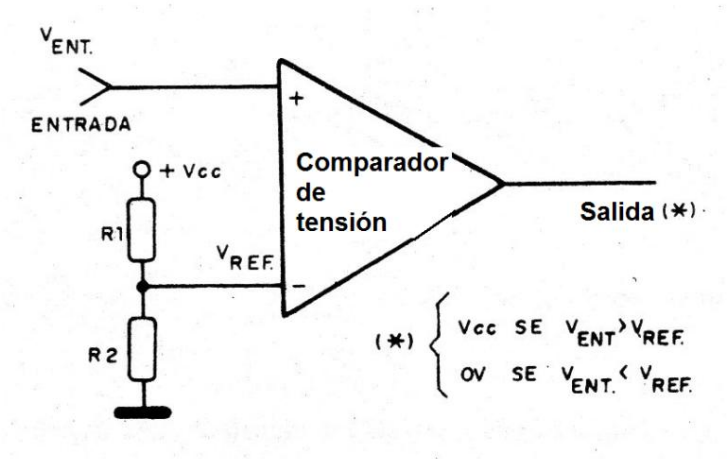

Figure 4: Example of a voltage comparator

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2.5. MOC3022 Optocoupler

The MOC3022 is an optocoupler with TRIAC output designed to control loads
alternating current without the need for a direct connection between the control part and the part
of power. It consists of an internal LED diode which, when activated, triggers a TRIAC
integrated.

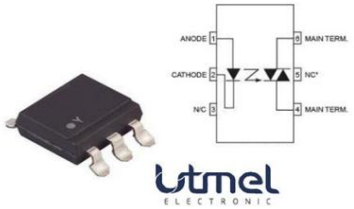

Figure 5: MOC3022 Optocoupler

2.6. TRIAC 2N6071

The 2N6071 is a TRIAC, a semiconductor device that allows current control
It alternates between both polarities of the AC waveform. It functions as a controlled switch.
by a small firing signal on its door

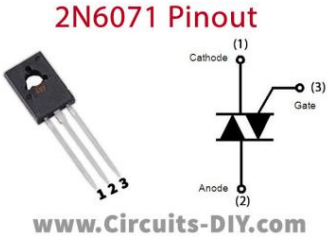

Figure 6: Triac 2N6071

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3. Development

For the circuit design, everything was divided into functional blocks, each with a specific task.
Clear. The first is the sound capture stage, which was done with a KY-038 microphone.
This module has two outputs, one digital and one analog; for this project the following was used:
It's analog because it delivers a signal proportional to the sound intensity. That signal
It is of very low amplitude, barely in millivolts, so a stage was needed
amplification to be able to process it properly.

3.1. Amplification with op amp (TL081)

Amplification was achieved using a TL081 in a non-inverting configuration, powered by
a symmetrical ±5V power supply. We take the +5V directly from the Arduino, and for the
At –5V we use an L7905 regulator with an external –9V source. In this stage also
A precision potentiometer was placed as a variable feedback resistor, for
adjust the amplifier's gain to control how loud a
applause to make it detected.

3.2. Active bandpass filter (3kHz – 6kHz)

Before choosing the frequencies for the filter, it was decided to use the Audacity program.
to analyze the frequencies present in applause, so an applause was recorded and
applied the frequency analysis tool.

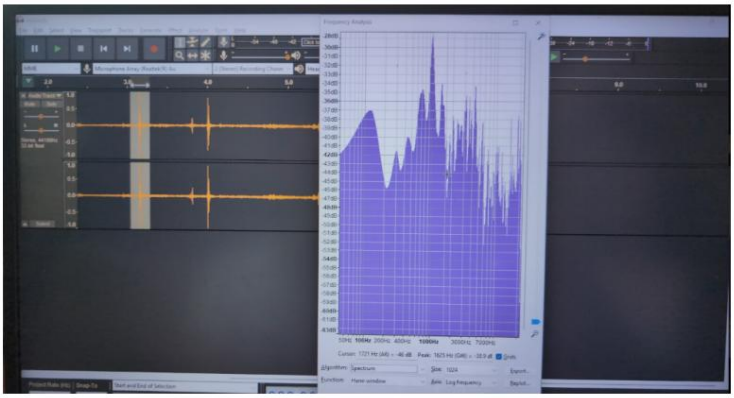

Figure 7: Frequency analysis in Audacity

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Seeing that a large part of the applause frequencies were applied in the same program, a filter with a curve simulating the one that the circuit filter will perform.

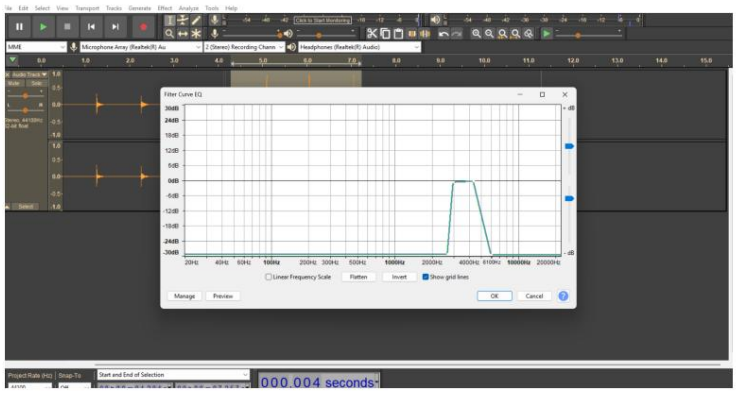

Figure 8: Filter simulation in Audacity

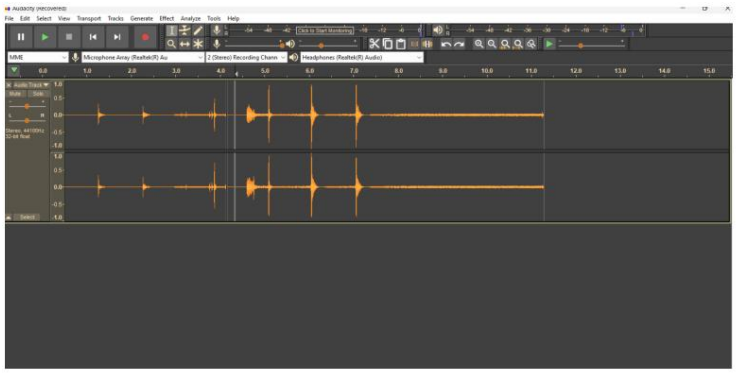

Figure 9: Filter output in Audacity

Once the filter is applied, it can be observed that the other background sounds are filtered out. of the audio and there are some peaks in the audio spectrogram, shown in figure 9 demonstrating that applause prevails after the filter.

Once frequencies for the filter were selected, the design of the filter began.

The capacitor was selected practically, choosing a value of 1nF since it is a an easy-to-obtain value. From that value, the necessary resistances were calculated for center the filter on the mid-frequency of:

$$fr = fH \cdot fL = \ddot{y}\ 6\ kHz \cdot 3\ kHz = 4.242\ kHz$$

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The interim band was defined as:

$$B = fH \ddot{y} fL = 6\ kHz \ddot{y} 3\ kHz = 3\ kHz$$

And the filter quality, such as:

$$Q = \frac{fr}{B} = \frac{4,242}{} = 1,414\ 3$$

With this data, and using the configuration of an active bandpass filter with amplifier
Operationally, the corresponding formulas were used to calculate the values of the
resistors, with the filter constant given by:

$$\frac{1}{RC} = 2\ddot{y}B$$

$$R = \frac{1}{2\ddot{y}BC} \ddot{y} \frac{0.1591}{1nF} \ddot{y}\ 53.033k\ddot{y}\ 3kHz \times$$

The following expression was used for the resistance

$$fr = 2 \frac{\sqrt{1 + \frac{R}{Rr}}}{\ddot{y}\ 2 \times \sqrt{\ddot{y}} \times R \times C}$$

$$Rr = R \left(\frac{fr \cdot RC}{0.1125} \right)^{2} \ddot{y}\ 1^{-1}$$

$$Rr \ddot{y}\ 17,684\ k\ddot{y}$$

3.3. Threshold comparator

After the filter, the signal was sent to a third stage: a voltage comparator made
also with a TL081. Here the filter signal was compared against a reference voltage
adjustable with a potentiometer. If the signal exceeded that threshold, the comparator saturated.
its high voltage (close to 5V), generating a digital output that represented the detection
from a round of applause. This output was connected directly to a digital input of the Arduino.

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One, who was in charge of processing the events. These 3 previous stages were simulated.

in Proteus 10 to observe its operation shown in figure 11.

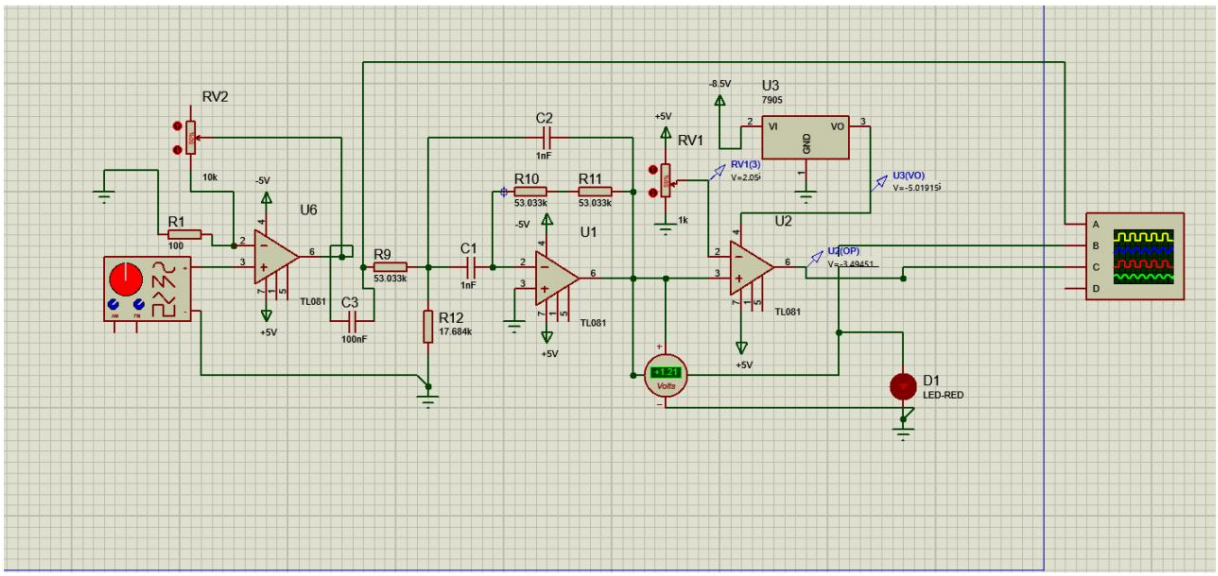

Figure 10: Frequency analysis in Audacity

3.4. Digital Processing

The Arduino was programmed to detect rising edges, that is, when the signal

It went from low to high. Upon detecting applause, it initiated a 3-second window to

count the following.

If there were 2 claps within that time, the light would turn on. If there were 4, it would turn off.

In the event that more than 3 seconds pass without reaching a valid amount or if a valid amount is detected

After more than 4 claps, the counter would automatically reset.

3.5. Power Stage

The final stage was triggering the 110V light bulb with a 2N6071 TRIAC. To isolate the

To control the mains voltage from the Arduino, an MOC3022 was used as an optocoupler.

It sent the signal to the internal LED of the MOC3022, and when it was activated, the TRIAC

It fired, allowing alternating current to pass to the light bulb.

All of this was integrated onto a single punched card with the three TL081s, the MOC3022, the

TRIAC and passive components, leaving the Arduino and the power supply connected by wires.

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4. Code

This was the code used, as explained above:

Listing 1: PID control code for the cart

```
const int pinEntrada = 2;                    // Comparator input
const int pinLED = 4;                        // LED (simulates a light bulb)

int applausecounter = 0;
unsigned long startTime = 0;
bool counting = false ;
bool lightOn = true ;

void setup() {
    pinMode ( pinEntrance ,      INPUT ) ;
    pinMode ( pinLED ,     OUTPUT );
    digitalWrite ( pinLED ,              HIGH);        // The light bulb starts on
    Serial . begin (115200) ;
    Serial. println ("Applause counter started... ");
}

void loop () {
    static bool previousState = LOW ;
    bool currentState = digitalRead(inputPin);

    // Detect rising edge
    if ( currentState == HIGH && previousState == LOW ) {
        if (! counting ) {
            startTime = millis();
            counting = true ;
        }

        applause counter ++;
        Serial.print ("Applause detected! Total: ");
        Serial. println (ApplauseCounter);

        // Immediate actions
```

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```
        if ( applausecounter == 2 && ! lightOn ) {
            digitalWrite ( pinLED ,          HIGH);
            lightOn = true ;
            Serial . println (" >> LIGHT ON <<") ;
        }
        else if ( applausecounter == 4 && lightOn ) {
            digitalWrite ( pinLED ,          LOW ) ;
            lightOn = false ;
            Serial. println (" >> FOCUS OFF <<") ;
        }
        else if (applausecounter > 4) {
            Serial. println (" >> More than 4 claps . Resetting counter . << " ) ;
            applause counter = 0;
            counting = false ;
        }

        delay (200) ; // Anti-bounce
    }

    previousState = currentState;

    // If the waiting time expires
    if ( counting && ( millis () - startTime >= 3000) ) {
        Serial . print (" >> Time expired . Counted applause : ") ;
        Serial. print ( applauseCounter ) ;
        Serial. println (". Resetting counter . <<") ;
        applause counter = 0;
        counting = false ;
    }
}
```

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5. Results

The circuit was initially simulated in Proteus, replacing the microphone with a signal generator. In this simulation the behavior was correct: the filter allowed the signal path in the desired range and attenuated out-of-band frequencies as it I expected. However, several details arose during the physical implementation that forced to modify the design. The KY-037 module used as a microphone introduced an offset of approximately 2.5V in the signal, which directly affected the operation of the filter. To mitigate this effect, a ceramic capacitor was incorporated at the input of the circuit in order to block that DC component, an adjustment that was also applied subsequently in the Proteus simulation to reflect the real behavior.

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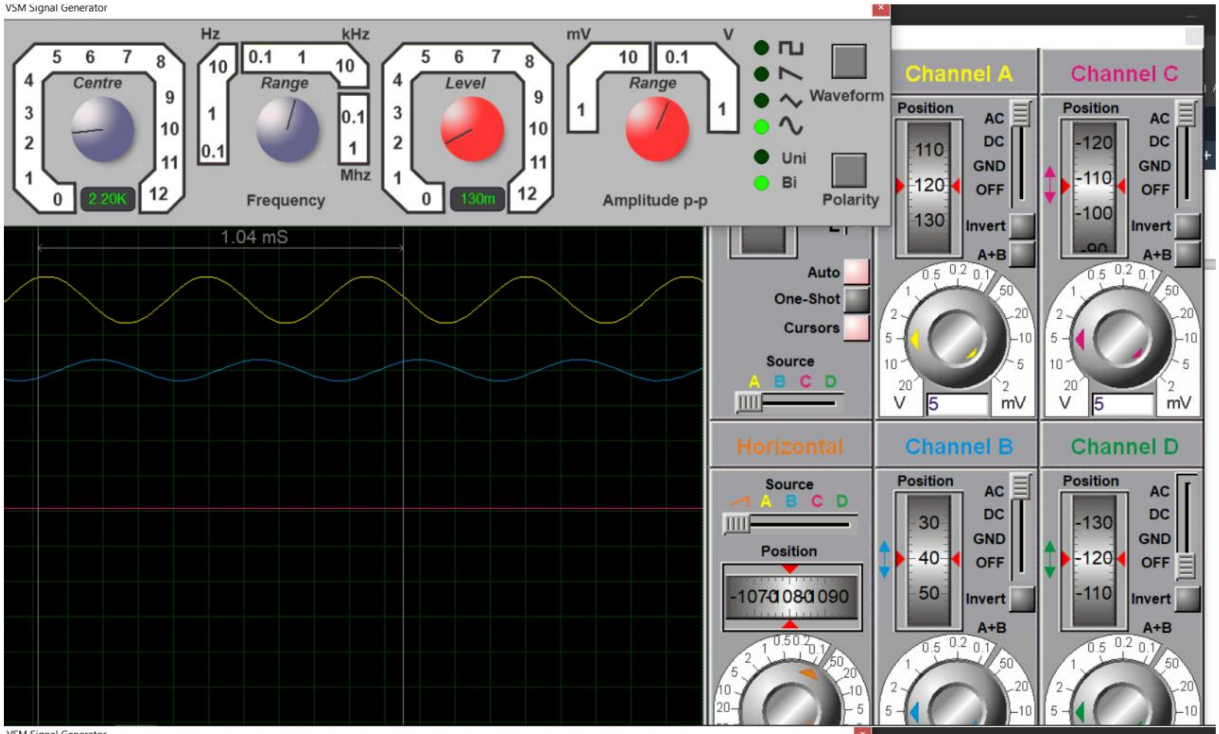

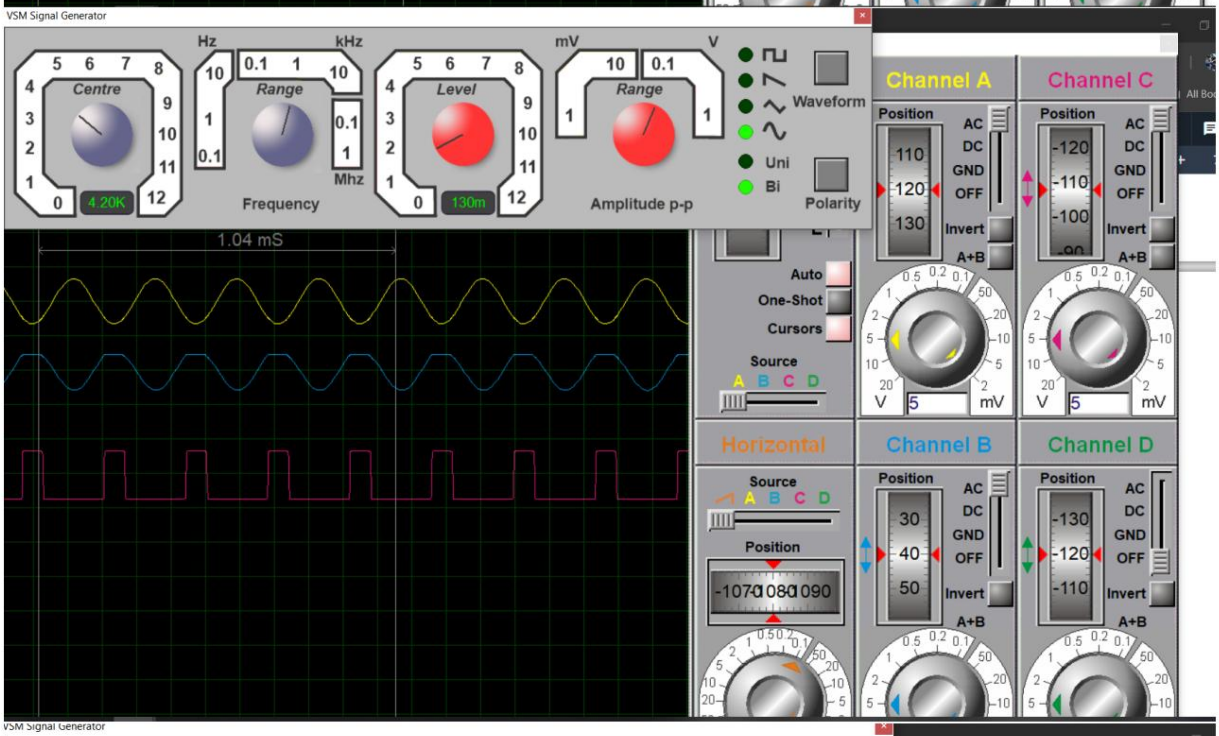

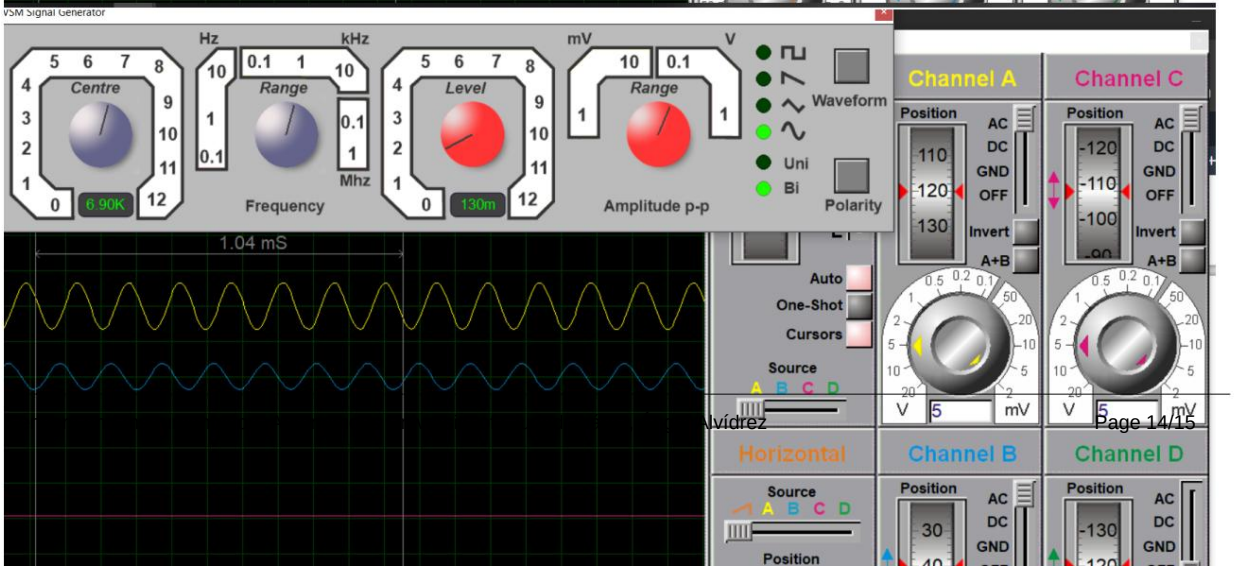

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Furthermore, it was detected that the comparator, in its low output state,
It introduced a small negative voltage that was undesirable at the time of excitation.
the TRIAC gate. This problem was solved by placing a diode that blocks that value
negative and ensures that the control signal remains clean and uninterrupted in the negative state
at rest. With these adjustments, both simulated and practical performance will be improved.
It was stable and the circuit fulfilled its objective of detecting acoustic signals within the
specific frequency range.

6. Conclusion

This project allowed the integration of multiple concepts of analog and digital electronics to
Develop a functional system for detecting specific frequencies using a filter
active bandpass filter with TL081 operational amplifiers and a trigger circuit with
Arduino-controlled TRIAC. Throughout the development process, its importance was verified.
to consider not only the theoretical design, but also the practical details that arise
when implementing real components, such as the unexpected offset of the KY-037 module or the
comparator behavior in low state.
Simulation in Proteus was a key tool for validating the design before the
physical weaponry, but it was also evident that certain elements do not behave in a
identical between the simulation and the practice. This forced direct adjustments to the
circuit, such as including a capacitor to eliminate DC offset or a diode to
Ensure signal levels compatible with TRIAC triggering. These simple solutions
But effective ones show how a good understanding of the behavior of each block is important.
Functional allows you to solve problems without redoing the entire design.
Finally, a system was achieved that detects signals in the desired range and responds
properly switching on an alternating current load, demonstrating practical use
of active filters, comparators and power devices in a control application
automated.

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Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 18 IoT Irrigation System

Sensors and Instrumentation

Angel Alvidrez 225581:

Date: May 21, 2025

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1. Introduction

Water scarcity and inefficient use of water resources are persistent problems.
in urban agriculture and gardening. In this context, automated irrigation systems
They represent an efficient solution for reducing water waste and improving care
of the plants.

The present project aims to develop an intelligent irrigation system using
an ESP32 development board, soil moisture sensors, and the Blynk platform for
Remote monitoring. This system allows real-time detection of humidity levels in the
ground and activate irrigation automatically when needed. In addition, the user can
View the plant's status from your mobile device, thanks to Wi-Fi connectivity
and the user-friendly interface offered by Blynk [1].

The idea was inspired by previous open-source projects, such as the one developed by
Viral Science [2], which showed a simple but functional approach to monitoring
plants through the integration of sensors and controllers connected to Blynk Starting
From this base, a customized solution was developed, adapted to the available resources.
and specific needs of the environment where it will be implemented.

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2. Theoretical Framework

The development of intelligent systems for monitoring and controlling plants has taken
of great relevance in the field of automation and the Internet of Things (IoT).
These systems allow for the optimization of resource use, such as water, improving efficiency and
facilitating plant care through the use of interconnected sensors and actuators.

2.1. ESP32 Microcontroller

The ESP32 is a dual-core microcontroller developed by Espressif Systems, which
It integrates Wi-Fi and Bluetooth connectivity, and offers high performance with low power consumption.
energetic [3]. Thanks to its versatility and processing capacity, it is widely
Used in IoT projects that require wireless communication and real-time control
real, like the smart irrigation system presented in this project.

Figure 1: ESP32 Development Board

2.2. Soil Moisture Sensor

The soil moisture sensor used measures the electrical conductivity between two
electrodes immersed in the substrate. The resistance varies depending on the content of
water, allowing the soil moisture level to be determined by means of an analog signal
that the microcontroller can interpret. This data is essential to activate the
water pump and maintain adequate irrigation.

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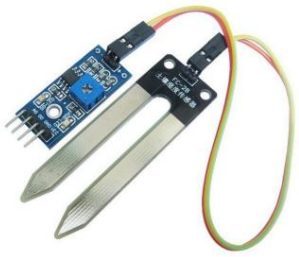

Figure 2: HR0040 Soil Moisture Sensor

2.3. DHT11 Sensor

The DHT11 sensor is a digital device that measures temperature and relative humidity.
of the air. It has a capacitive humidity sensor and a thermistor for temperature,
delivering values in digital format with sufficient resolution for applications
basic environmental conditions. In the system, this sensor allows monitoring of atmospheric conditions.
that influence the state of the plant.

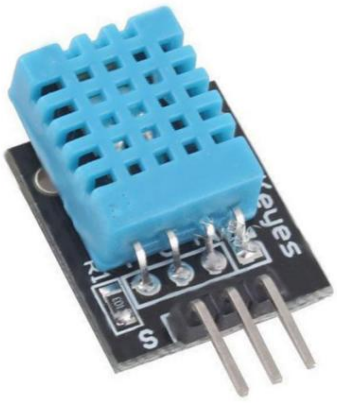

Figure 3: Air humidity and temperature sensor

2.4. PIR Sensor (Motion Sensor)

The PIR (Passive Infrared) sensor detects movement based on changes in the
Infrared radiation emitted by objects in its field of view. It is used to detect
the presence or passage of people or animals near the plant, enabling alerts and
reports in the mobile application when the sensor is active.

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Figure 4: PIR motion sensor

2.5. 5V Relay Module

The 5V relay module allows you to control AC or DC devices
continuous higher voltage and current, such as the water pump, using the signal of
Low voltage generated by the microcontroller. This module acts as a switch
electronic shielding that isolates the microcontroller from high electrical loads, ensuring the
system security.

Figure 5: 5V relay module

2.6. Water Pump and Flow Control

The water pump used is a 5V DC pump, designed for
supply a constant flow of water suitable for watering small plants and
pots. The key feature of this pump is its nominal flow rate of approximately 1.6
liters per minute [4] (0.0267 liters per second), which allows for precise calculation of the
Total water consumption during operation. The pump is activated by

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a 5V relay module that functions as an electronic switch, controlled directly
by the ESP32 to turn the irrigation on or off according to the conditions detected by the
sensors.

Precise control of water flow is essential to avoid both insufficient and excessive irrigation.
excess, optimizing the use of water resources and promoting healthy development of
the plants. In addition, the system calculates and displays the accumulated consumption in real time.
of water during pump operation, facilitating efficient monitoring.

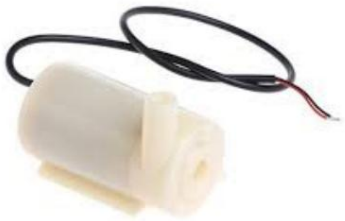

Figure 6: 9V Water Pump

2.7. Blynk Platform

Blynk is an IoT platform that allows the rapid creation of graphical interfaces for the
Remote control and monitoring of embedded devices. Through its mobile application, the
Users can view data, receive notifications, and control actuators in real time
real through customizable widgets [1]. In this project, Blynk facilitates monitoring
of the environmental conditions and the remote management of the irrigation system.

2.8. LCD Screen

The system uses a 16x2 character alphanumeric LCD screen, which is based
on the standard HD44780 controller. This screen allows important information to be displayed.
such as temperature, humidity, soil moisture, and water consumption in a clear and
user-friendly.

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Figure 7: 16x2 LCD Screen

2.9. I2C Communication and PCF8574 Module

To simplify the connection between the ESP32 microcontroller and the LCD screen, the following is used:

An I2C adapter module based on the PCF8574 chip. This module reduces the number of Pins needed to control the screen, using only two data lines, SDA and SCL.

The PCF8574 chip acts as a port expander that allows control of the pins of the LCD via the I2C protocol, facilitating communication and reducing complexity of the wiring in the project.

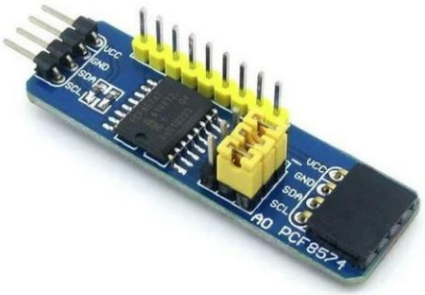

Figure 8: PCF Module

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3. Development

The smart irrigation system was developed using an ESP32 microcontroller and sensors.
environmental sensors, a relay module, and an LCD screen for data display.
The main elements and the implemented operating logic are described below.

3.1. Microcontroller and communication

An ESP32 was used due to its ability to connect to WiFi networks and run the
Blynk platform, which allows remote monitoring and control of the system through a
Mobile application. Communication with sensors and actuators is done via pins
digital and analog, as well as via the I2C bus for the LCD screen.

3.2. Sensors and actuators

The system integrates the following devices:

- Soil moisture sensor connected to analog pin GPIO36, which measures the percentage of moisture present.
- DHT11 temperature and humidity sensor connected to GPIO4, which provides environmental conditions.
- PIR sensor connected to GPIO27, detects movement for additional functions monitoring.
- 5V relay module connected to GPIO26, which controls the power on/off the water pump.
- Physical button on GPIO33 to manually activate or deactivate the relay.

3.3. Program Logic

The code uses the Blynk library to connect the device to the cloud and enable the
Remote interaction. It also uses timers to periodically read the sensors.
and update the LCD screen.

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The water pump is activated or deactivated depending on the state of the physical or virtual button. During Its operation calculates the approximate water consumption based on the time that It remains on and has an estimated flow of 1.6 liters per minute.

3.4. Data Visualization

The LCD screen displays real-time temperature, ambient humidity, and humidity of the soil and accumulated water consumption. The information is updated every second. to maintain a clear and useful user interface.

3.5. System Operation

The system initiates the WiFi connection and synchronizes with the Blynk platform. Configuration begins. the pins and sensors, and periodic tasks are set up using timers for reading and updating data.
In the main loop, the state of the PIR sensor is evaluated to activate indicators of During movement, Blynk's tasks are executed and user interactions are monitored. through physical and virtual buttons.
This modular architecture allows the system to be flexible, efficient, and scalable for future improvements.

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4. Code

This was the code used, as explained above:

Listing 1: PID Speed Code

```
// Viral Science www . viralsciencecreativity. com www. youtube. com /c/ viralscience
// Blynk IOT Smart Plant Monitoring System

# define BLYNK_TEMPLATE_ID " TMPL2I5BNgZpU "
# define BLYNK_TEMPLATE_NAME " Intelligent Irrigation System "
# define BLYNK_AUTH_TOKEN " sP_bnZb903gxCO6MNO - h4YMteKdv925c "

# include < Wire .h >
# include < LiquidCrystal_I2C .h >
# include < WiFi .h >
# include < WiFiClient .h >
# include < BlynkSimpleEsp32 .h >
# include < DHT .h >

# define BLYNK_PRINT Serial

// LCD
LiquidCrystal_I2C lcd (0 x20 , 16 , 2) ;

// I2C Pins
const int SDA_PIN = 21;
const int SCL_PIN = 22;

// WiFi and Blynk
char auth [] = BLYNK_AUTH_TOKEN ;
char ssid[] = " ********** ";
char pass [] = " ******** ";

// Blynk Timer
BlynkTimer timer;

// Pins
```

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```
# define SOIL_SENSOR_PIN 36
# define PIR_PIN 27
# define RELAY_PIN_1 26
# define PUSH_BUTTON_1_PIN 33
# define VPIN_BUTTON_1 V12
# define VPIN_CONSUMO V13

// DHT11
# define DHTPIN 4
# define DHTTYPE DHT11
DHT dht ( DHTPIN , DHTTYPE ) ;

// States
int PIR_ToggleValue = 0;
int relay1State = LOW ;
int pushButton1State = HIGH ;

// Consumption
unsigned long onTime = 0; // in seconds
float consumptionLiters = 0.0;
const float LITERS_PER_SECOND = 1.6 / 60.0; // 0.0267

void setup() {
    Wire.begin(SDA_PIN) , SCL_PIN ) ;
    Serial . begin (9600) ;

    lcd . begin (16 , 2) ;
    lcd . backlight () ;
    lcd . setCursor (0 , 0) ;
    lcd. print ("Initializing ");
    for ( int a = 5; a <= 10; a++) {
        lcd.setCursor(a , 1) ;
        lcd . print (".") ;
        delay (500) ;
    }
    lcd . clear () ;
```

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```
    pinMode ( PIR_PIN , INPUT ) ;
    pinMode ( RELAY_PIN_1 , OUTPUT ) ;
    digitalWrite ( RELAY_PIN_1 , relay1State ) ;
    pinMode ( PUSH_BUTTON_1_PIN , INPUT_PULLUP ) ;

    dht . begin () ;
    Blynk.begin(auth , ssid, pass, " blynk . cloud ", 80) ;

    timer . setInterval (100 L , soilMoistureSensor ) ;
    timer . setInterval (2000 L , DHT11sensor ) ;
    timer . setInterval (500 L, checkPhysicalButton);
    timer . setInterval (1000 L , calculateConsumption ) ; // every second
}

// Soil moisture sensor reading
void soilMoistureSensor() {
    float value = analogRead ( SOIL_SENSOR_PIN ) ;
    value = map ( value , 0 , 4095 , 0 , 100) ;
    value = ( value - 100) * -1;
    Blynk . virtualWrite ( V3 , value);
    lcd . setCursor (0 , 1) ;
    lcd . print ("S:") ;
    lcd.print(value);
    lcd . print (" ") ;
    Serial . print (" SOIL MOISTURE: ") ;
    Serial . println ( value ) ;
}

// DHT11 sensor reading
void DHT11sensor () {
    float h = dht . readHumidity () ;
    float t = dht . readTemperature () ;
    if ( isnan ( h ) || isnan ( t ) ) {
        Serial . println (" Error reading DHT11 ") ;
        return ;
    }
```

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```
    Blynk. virtualWrite ( V0 Blynk .        ,   t) ;
    virtualWrite ( V1                       ,   h) ;

    lcd . setCursor (0          ,   0) ;
    lcd . print ("T:") ;
    lcd . print ( t ) ;
    lcd . print ("C ") ;
    lcd . print ("H:") ;
    lcd . print ( h ) ;
    lcd . print (" % ") ;
    Serial. print ("Temp: "); Serial. print ( t ) ;
    Serial. print ("C Hum: "); Serial. print ( h ) ; Serial. println ("%") ;
}

// PIR sensor
void PIRsensor() {
    bool value = digitalRead ( PIR_PIN ) ;
    Serial. print ("PIR Sensor Value: ") ;
    Blynk . virtualWrite ( V5               ,   value);
    Serial . println ( value ) ;

    if ( value ) {
        Blynk. logEvent ( " pirmotion ", " WARNING ! Motion Detected !") ;
        WidgetLED LED ( V5 ) ;
        LED . on () ;
    } else {
        WidgetLED LED ( V5 ) ;
        LED . off () ;
    }
}

// Physical B
void checkPhysicalButton() {
    if ( digitalRead ( PUSH_BUTTON_1_PIN ) == LOW ) {
        if ( pushButton1State != LOW ) {
            relay1State = ! relay1State ;
            digitalWrite ( RELAY_PIN_1 , relay1State ) ;
```

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```
            Blynk. virtualWrite ( VPIN_BUTTON_1 , relay1State ) ; delay (500) ;


        }
        pushButton1State = LOW ;
    } else {
        pushButton1State = HIGH ;
    }
}

// PIR status from Blynk
BLYNK_WRITE ( V6 ) {
    PIR_ToggleValue = param . asInt () ;
}

// Virtual button connected to relay
BLYNK_CONNECTED () {
    Blynk. syncVirtual ( VPIN_BUTTON_1 ) ;
}

BLYNK_WRITE ( VPIN_BUTTON_1 ) {
    relay1State = param . asInt () ;
    digitalWrite ( RELAY_PIN_1 , relay1State ) ;
}

// Function to calculate water consumption
void calculateConsumption() {
    if ( relay1State == HIGH ) {
        timeOn ++;
        consumptionLiters = ignitionTime * LITERS_PER_SECOND ;
        consumptionLiters = round ( consumptionLiters * 100.0) / 100.0; // round to 2 decimals
    }

    Blynk. virtualWrite (VPIN_CONSUMPTION    ,    consumptionLiters ) ;


    lcd . setCursor (11              ,    1) ;
    lcd . print ("L:") ;
    lcd. print ( consumptionLiters ) ;
```

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```
    lcd.print (" ); // clear leftover text

    Serial.print ("WATER CONSUMPTION: ");
    Serial. print ( consumptionLiters ) ;
    Serial . println (" L") ;
}

// Main loop
void loop () {
    if ( PIR_ToggleValue == 1) {
        lcd . setCursor (5          ,   1) ;
        lcd . print ("M:ON ") ;
        PIRsensor() ;
    } else {
        lcd . setCursor (5          ,   1) ;
        lcd. print ("M:OFF ") ; WidgetLED
        LED (V6);
        LED . off () ;
    }

    Blynk . run () ;
    timer.run();
}
```

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5. Results

During the implementation and testing of the smart irrigation system, it was possible to monitor real-time soil moisture, ambient temperature and humidity, as well as approximate water consumption based on pump operating time.

The consumption calculation was based on an estimated flow of 1.6 liters per minute, obtained from technical specifications of the pump. However, for more accurate measurements, It would have been ideal to use a flow meter that would allow recording the actual flow of water. However, flow meters are often expensive devices to integrate into projects. due to the low budget, the indirect method based on the following was chosen at this stage: the relay's operating time.

The data obtained showed an adequate response of the system to the conditions of the soil and environment, with the ability to manually activate or deactivate the pump The LCD screen provided clear and continuous information to the user, facilitating system status monitoring without the need for a remote connection.

Overall, the results confirm that the system meets the stated objectives. for efficient irrigation monitoring and control, with room for future improvements in accuracy of water consumption.

6. Conclusion

The development of the intelligent irrigation system based on ESP32 proved to be a solution effective for the automated monitoring and control of soil conditions and environment, optimizing water use through precise pump activation.

The use of sensors such as the DHT11 and the soil moisture sensor allowed us to obtain data reliable devices that were successfully integrated with the Blynk platform for monitoring remote control, while local control via a relay and a physical button increased the system versatility.

Although the calculation of water consumption was approximate due to the absence of a For the flow meter, the method based on operating time proved sufficient to estimate the volume used, while maintaining affordable costs.

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The implementation of the LCD screen and the PCF module facilitated direct interaction with the user with the system, improving the experience and on-site control.

Finally, the project opens the door for future improvements, such as the incorporation of More precise sensors and the integration of intelligent algorithms for further optimization irrigation, contributing to a sustainable use of water resources.

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Publications and Academic Works

Published Article: Kinematic Modelling of Five-Bar Bipedal Locomotion

Kinematic Modelling of Five-Bar Bipedal Locomotion: An Analytical Approach

Status: Published
Publication: Proceedings of the 30th International Conference on Methods and Models in Automation and Robotics (MMAR 2026)
Year: 2026 — Pages: 64–69

Authors: Ángel H. Alvídrez-Hernández, Edgar A. Martínez-García

DOI: 10.1109/MMAR70562.2026.11667841
IEEE Xplore: View Publication

Description:
This work presents an analytical model for the kinematics of a bipedal robot based on a five-bar mechanism. Direct and inverse kinematics equations are developed, along with a deterministic mapping between joint space and task space. Furthermore, a Kalman filter is implemented for system state estimation, enabling trajectory tracking and motion stabilization in an experimental prototype.

Kinematic Modelling of Five-Bar Bipedal Locomotion: An Analytical Approach

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***Abstract*—This paper presents an analytical framework for the kinematic modeling of a five-bar bipedal robot, which is a home-made experimental prototype. The five-bar linkage introduces complex holonomic constraints and closed-loop dependencies that require precise mathematical formulation. We derive the forward and inverse kinematic equations using an analytical approach, providing a deterministic solution for the foot trajectory relative to the hip base. The study further establishes a closed-form mapping between joint space and task space, this research provides as a first approach for path tracking and control a Kalman filter for straight trajectories. The proposed model is validated through numerical simulations and laboratory experiments, characterizing the workspace and motion primitives of five-bar bipedal locomotion.**

***Index Terms*—Bipedal robot, five-bar linkage, kinematic-motion, Kalman-filter, Walking-gait**

I. Introduction

Modern robotics shifts from heavy industrial arms to agile, mobile agents, where these systems are more relevant than ever. While massive humanoid robots often grab headlines, five-bar bipedal systems represent efficiency and simplicity. Five-bar limb mechanisms are able to control a foot's position using servomotors located near the hip, consequently reducing swing mass effects, making lighter and more responsive limbs (home-made experimental prototype Fig. 1ab). Unlike wheeled robots, bipeds navigate complex obstacles such as curbs or stairs. Their closed-loop nature makes them more rigid and energy-efficient than traditional serial-link robots.

A purpose of this research is to provide a physics-based bridge between theoretical modeling and physical experimentation. This approach is tied to holonomic constraints and closed-loop dependencies, which are non trivial issues. A five-bar system is a parallel multi-bar linkage that offers high payload-to-weight ratio (Fig. 1c). The structure shares the mechanical load across multiple bars. Besides, performs high precision as metric errors in one joint do not necessarily becomes as so accumulative as they do in serial multi-legged robots. This kind of robotic platform (Fig.1) provides a way to understand its inverse kinematics, both the joints positioning and the whole body displacement. The implementation of a Kalman filter is a suitable technique for sensor fusion in order to deal with noisy sensor data due to mechanical vibrations in real-time. In principle, research into motion primitives allows global motion planning by characterizing the limbs' workspace, and use it to perform multiple robot's gaits suitable for different terrains. A main summarized contribution of this paper is to provide a deterministic kinematic solution that allows for walking path tracking. Moreover, dynamic balance by using the Kalman Filter to estimate the robot's state and prevent it from tipping over. Essentially, the five-bar bipedal robot is deployed to solve the problems of legged locomotion.

This manuscript is organized as follows: Section II provides a comparative analysis of related work. Section III introduces the robot's architecture and mechanical design. Section IV derives the forward and inverse kinematics of the five-bar linkage. Section V presents the algebraic deduction for the robot's multi-part kinematic structure. Section VI details the implementation of the Kalman filter for motion estimation. Finally, Section VII offers concluding remarks.

II. Related Work

The design and performance characterization of underactuated bipedal systems have been extensively documented, notably in [1], while more recent efforts have employed derivative-free optimization to refine leg parameters for ZMP estimation and energetic efficiency [2]. Within this domain, five-bar mechanisms have emerged as a versatile solution, facilitating bio-inspired motion [3], high-agility maneuvering [4], and the exploration of fundamental kinematic properties [5]. To address the inherent instabilities of legged locomotion, researchers have further implemented inverted pendulum control strategies to regulate tilt during high-speed gait [6]. Beyond traditional bipedalism, multi-bar linkages have found diverse applications—serving as sophisticated suspension systems for self-balancing platforms [7], [8] and as bio-mimetic propulsion for underwater robotics [9], [10]. Similarly, the Klann linkage continues to represent a foundational framework for quadrupedal locomotion across a wide spectrum of experimental and industrial paradigms [11]–[14].

III. Robot Structure

This section details the robotic platform design [15], encompassing its mechanical structure, computational architecture, instrumentation, and sensor-actuator integration. Figure 1ab illustrates the bipedal experimental prototype alongside its

CAD model. Additionally, figure 1c clarifies the limb's linkage arrangement, demonstrating its geometric consistency with the isolated five-bar mechanism shown in Fig. 3.

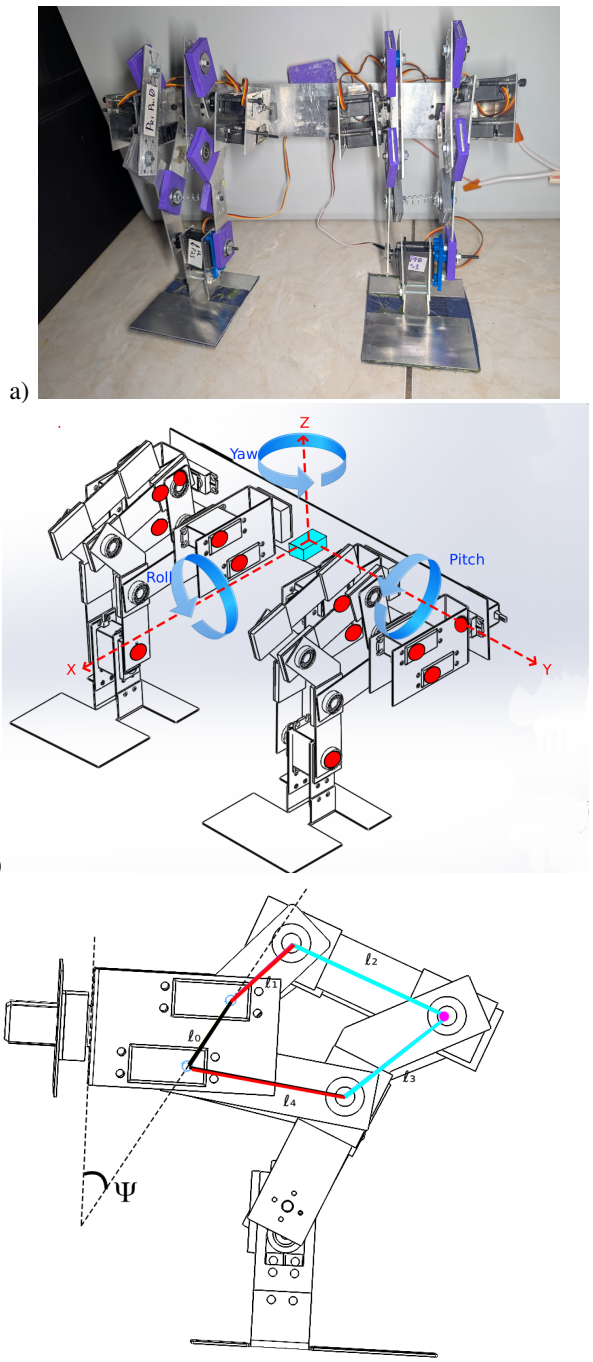

Fig. 1. Robotic bipedal five-bar linkage limbs. a) Experimental prototype. b) Detailed CAD model showing motor-potentiometer placement (red points) and central IMU (blue rectangle). c) Limb's side view.

The robotic architecture, shown in Figure 2, comprises an Orange Pi Zero (Debian Linux) interfacing with an ESP32 slave module for low-level control of twelve servomotors. The instrumentation includes a 9-axis IMU located at the center of mass and twelve potentiometers used as angular transducers to monitor the roll and pitch of each limb.

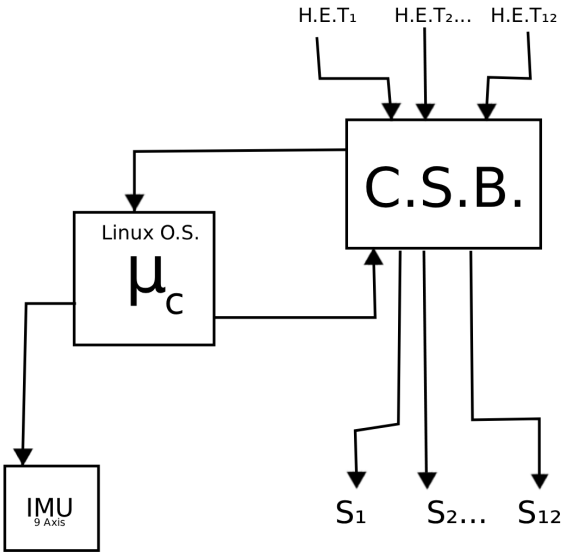

Fig. 2. The master unit (μ_c) interfaces with the Control and Sensing Board (C.S.B.) to manage an array of twelve Hall Effect Transducers (H.E.T. $_1$ to H.E.T.$_{12}$) and twelve servos (S_1 to S_{12}).

IV. Limb's Kinematics

A. Forward Kinematics

The kinematics development of a Five-Bar Bipedal locomotion is based on the analysis of one closed-chain 5-bar mechanism. This analysis will be divided into 2 main stages, first the forward kinematics (Fig.3a), which allows us to determine the position of our end-effector from the actuator angles, and the inverse kinematics will allow us to determine the angles of our actuators from desired coordinates for our end-effector (Fig.3b), alternative approach in [16]. We begin by defining our X and Y coordinates from the end of ℓ_1 and ℓ_4 with our following expressions:

$$x_1 = \ell_1 \cos(\theta_1); \quad y_1 = \ell_1 \sin(\theta_1); \tag{1}$$

$$x_4 = \ell_0 + \ell_4 \cos(\theta_4); \quad y_4 = \ell_4 \sin(\theta_4). \tag{2}$$

These positions are necessary to know the instant distance from the ends of ℓ_1 and ℓ_4 which we call ℓ_e and is defined geometrically by:

$$\ell_e = \sqrt{(x_1 - x_4)^2 + (y_1 - y_4)^2} \tag{3}$$

ℓ_e is a measurement from the inter-geometric relation between the angles θ_1 and θ_4, if the following condition is true;

$$(\ell_2 + \ell_3) < \ell_e \tag{4}$$

Then, it is known that the closed kinematic chain cannot reach the desired posture. Thus, to determine inner angles β, α and γ, the sine and cosine rules obtained the expressions:

$$\beta = \arccos\left(\frac{{\ell_e}^2 - {\ell_2}^2 - {\ell_3}^2}{-2\ell_2\ell_3}\right), \tag{5}$$

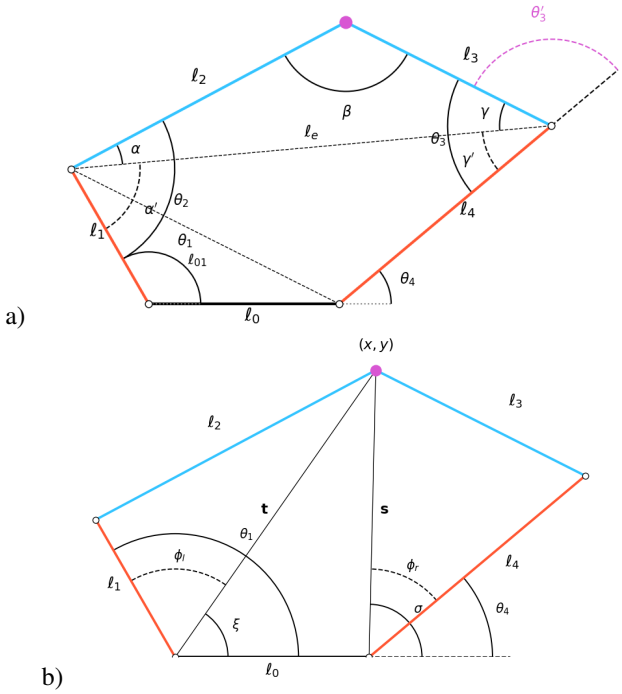

Fig. 3. Limb's 5-bar linkage geometric model. a) Forward kinematics. b) Backward kinematics.

$$\alpha = \arcsin(\frac{\ell_3}{\ell_e}\sin(\beta)), \tag{6}$$

$$\gamma = \arcsin(\frac{\ell_2}{\ell_e}\sin(\beta)). \tag{7}$$

To continue we require the local version of the γ, which we'll call γ'. However, to calculate γ' we must first determine the instantaneous distance from the top end of ℓ_1 and the right end of ℓ_0 as seen on Figure 3. By applying the law of cosines, we obtain the expression for this distance, ℓ_{01}:

$$\ell_{01} = \sqrt{{\ell_0}^2 + {\ell_1}^2 - 2\ell_1\ell_0\cos(\theta_1)}, \tag{8}$$

which we use to calculate γ' using the expression:

$$\gamma' = \arccos\left(\frac{{\ell_{01}}^2 - {\ell_4}^2 - {\ell_e}^2}{-2\ell_e\ell_4}\right) \tag{9}$$

As observed in geometric scheme in Fig.3, global angle θ_3 and local angle θ_3' are supplementary, as they lie on the same horizontal axis. Hence, their relationship is defined as:

$$\theta_3' = \pi - \theta_3, \tag{10}$$

which is needed to calculate the coordinates of our end-effector using:

$$p_x = \ell_0 + \ell_4\cos(\theta_4) + \ell_3\cos(\theta_4 + {\theta_3}'), \tag{11}$$

$$p_y = \ell_4\sin(\theta_4) + \ell_3\sin(\theta_4 + {\theta_3}'). \tag{12}$$

B. Inverse Kinematics

Conversely, inverse kinematics aims to determine the active joint angles, θ_1 and θ_4 required to position our end-effector at known Cartesian coordinates. By defining the position $\mathbf{t}$, pointing out to the end-effector, its magnitude and true elevation angle relative to the local horizontal ground plane are determined.

$$||\mathbf{t}|| = \sqrt{p_x^2 + p_y^2}, \tag{13}$$

$$\xi = \arctan\left(\frac{p_y}{p_x}\right). \tag{14}$$

Thus, to find the angle between ℓ_1 and $\mathbf{t}$, the cosine rule yields:

$$\phi_l = \arccos\left(\frac{\ell_2^2 - ||\mathbf{t}||^2 - \ell_1^2}{-2\ell_1||\mathbf{t}||}\right) \tag{15}$$

And the angular position of the left actuator is defined by:

$$\theta_1 = \xi + \phi_l \tag{16}$$

Let us continue the analysis by getting the angular position of the right actuator, which is calculated by defining a positional vector defined as $\mathbf{s}$ and its magnitude and elevation with respect to the ground come from the equations:

$$||\mathbf{s}|| = \sqrt{(p_x - \ell_0)^2 + p_y^2} \tag{17}$$

$$\sigma = \arctan\left(\frac{p_y}{p_x - \ell_0}\right) \tag{18}$$

The angle between ℓ_4 and $\mathbf{s}$ is a cosine rule such that,

$$\phi_r = \arccos\left(\frac{\ell_3^2 - ||\mathbf{s}||^2 - \ell_4^2}{-2||\mathbf{s}||\ell_4}\right). \tag{19}$$

Due to the geometric configuration of the right side, the internal angle is subtracted from σ, so the angular position of the right actuator is defined by:

$$\theta_4 = \sigma - \phi_r \tag{20}$$

According to Fig. 4, it is a track of a square-shape trajectory to validate the inverse kinematics accuracy, demonstrating precise path tracking and correct execution of sharp directional changes.

V. Biped's Kinematics

Due to the fact that the legs of the platform do not operate on a static plane, it is necessary to transform the local system coordinates to a global 3D reference system, with its origin located on the center of its hip. This transformation consists of 3 fundamental rotations, a translation defined by:

$$\mathbf{R_z}\left(-\frac{\pi}{2}\right) = \begin{pmatrix} \cos\left(\frac{\pi}{2}\right) & -\sin\left(-\frac{\pi}{2}\right) & 0 \\ \sin\left(-\frac{\pi}{2}\right) & \cos\left(-\frac{\pi}{2}\right) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{21}$$

$$\mathbf{R_y}\left(\frac{\pi}{2}\right) = \begin{pmatrix} \cos\left(\frac{\pi}{2}\right) & 0 & \sin\left(\frac{\pi}{2}\right) \\ 0 & 1 & 0 \\ -\sin\left(\frac{\pi}{2}\right) & 0 & \cos\left(\frac{\pi}{2}\right) \end{pmatrix}. \tag{22}$$

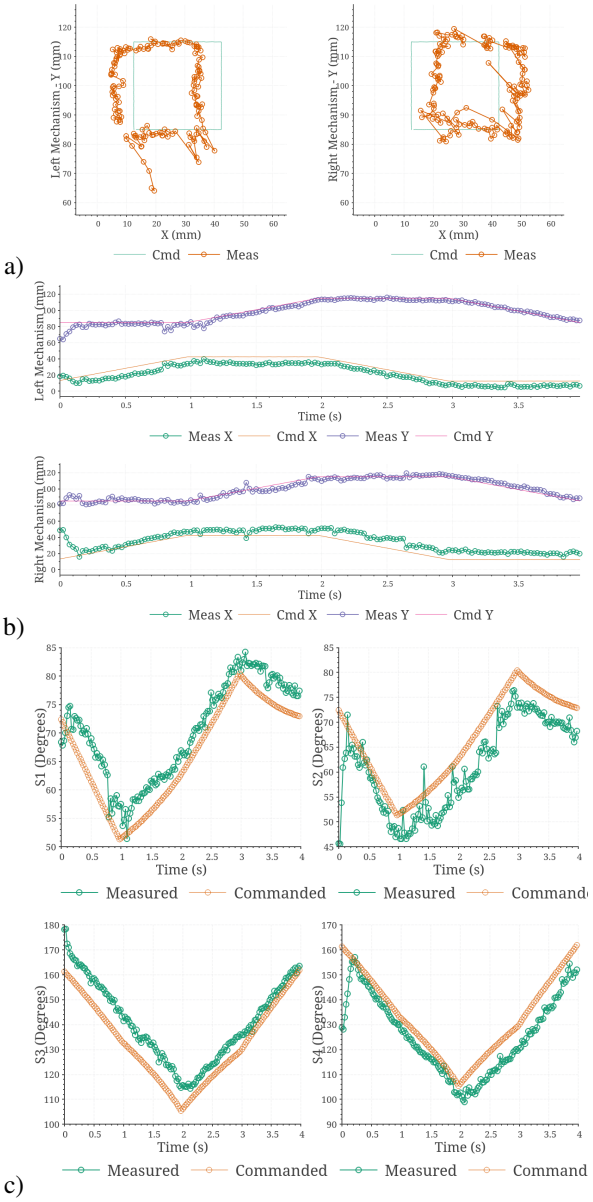

Fig. 4. Limb kinematics validation for a square trajectory. a) Spatial workspace tracking in task space. b) End-effector displacement over time. c) Active joints angular commands and measurements.

The leg rotation is the angle between ℓ_0 and the hip, set as ψ,

$$\mathbf{R_z}(\psi) = \begin{pmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{23}$$

The final rotation matrix is multiplying the individual rotation matrices, giving us the expression:

$$\mathbf{R_{final}} = \mathbf{R_z}\left(-\frac{\pi}{2}\right)\mathbf{R_y}\left(\frac{\pi}{2}\right)\mathbf{R_z}(\psi) \tag{24}$$

Finally, the three-dimensional position $\mathbf{p_{3D}}$ is given by applying the rotation to the local vector and adding the translation vector, which is obtained from the offset on the y axis;

$$\mathbf{p_{3D}} = \mathbf{R_{final}}\mathbf{p_{local}} + \begin{pmatrix} 0 \\ y_{offset} \\ 0 \end{pmatrix} \tag{25}$$

Now knowing the limbs and biped's kinematics we shall continue with the workspace of both. The limbs workspace is evaluated sweeping both actuators angles θ_1 and θ_4 across their physical limits $[0, \pi]$ while also filtering out unreachable coordinates by enforcing the geometric constraint indicated on Equation 4 Subsequently, the 3D workspace of the bipedal

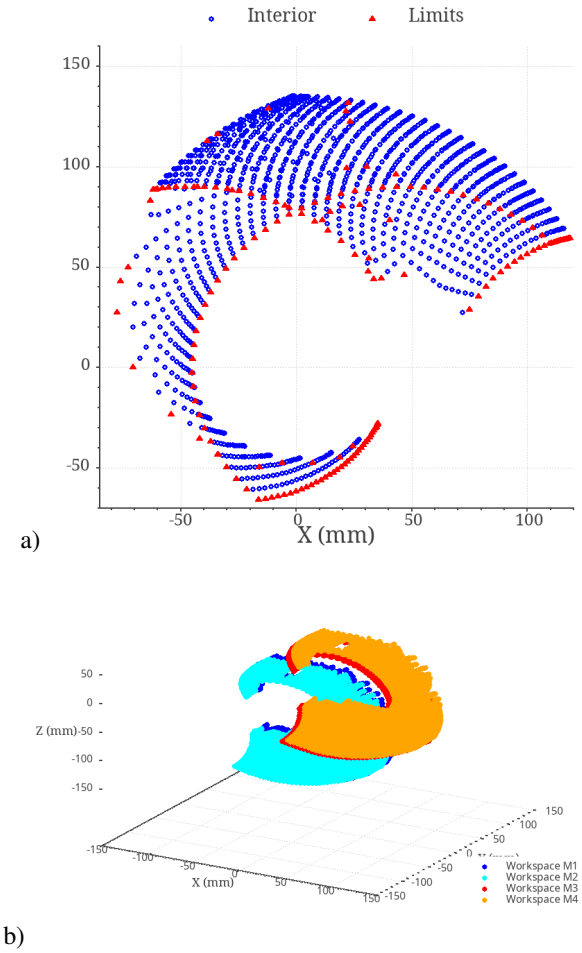

Fig. 5. Bipedal gaits workspace. a) Planar; and b $_c$) volumetric.

platform is computed by mapping the 2D boundaries into the spatial domain. Applying the rotation and offset of each leg to the planar workspace we add an extra rotation which will allow the legs to do abduction and adduction movements with limits of $[-15\frac{\pi}{180}, 15\frac{\pi}{180}]$ and doing a sweep of all the active articulations giving us a comprehensive mapping of the biped's total reachable volume, which is crucial for trajectory planning and gait generation. The bipedal locomotion is achieved by driving the active joints with harmonic continuous functions.

Let $\phi_1(t)$ and $\phi_4(t)$ be the angular positions for the primary and secondary actuators of the right leg, defined as:

$$\phi_{1_R}(t) = \phi_{1_{base}} + A\sin(\omega t),$$

$$\phi_{4_R}(t) = \phi_{4_{base}} + A\sin\left(\omega t + \frac{\pi}{2}\right),$$

where ϕ_{base} denotes the central angular offset, A is the amplitude, and ω the angular frequency of the gait cycle. To achieve an alternating walking pattern, the left leg actuators operate with a phase shift of π radians relative to the right leg, ensuring that while one leg is in the swing phase, the other remains in the stance phase:

$$\phi_{1_L}(t) = \phi_{1_{base}} + A\sin(\omega t + \pi),$$

$$\phi_{4_L}(t) = \phi_{4_{base}} + A\sin\left(\omega t + \frac{3\pi}{2}\right).$$

Finally, the 3D spatial trajectory of each foot, $\mathbf{P}_{foot}(t)$, is computed by mapping these time-varying joint angles through the forward kinematics function $\mathbf{FK}$ and applying the hip's rotational matrix $\mathbf{R}_{final}$ and translation vector $\mathbf{T}_{offset}$:

$$\mathbf{P}_{foot}(t) = \mathbf{R}_{final} \cdot \mathbf{FK}(\phi_1(t), \phi_4(t)) + \mathbf{T}_{offset}.$$

Figure 6b shows both limbs synchronized gaits, and such cycloids are ones of the so numerous existing in the set of the workspace (Fig.5). Figure 6a includes the input angles in active joints to yield the cycloid.

VI. Motion State Estimation

In a first approach, to ensure stable locomotion along straight trajectories, accurate estimation of the robot's position and speed is implemented. Sensor data inherently contain noise, which can destabilize the gait cycle. To reduce this, a Kalman filter was implemented for path tracking. The motion of the biped is modeled as a linear dynamic system. The state vector $\mathbf{x_k} \in \mathbb{R}^4$ at time step k contains the 2D position and velocity of the robot, defined as $\mathbf{x_k} = [x, y, \dot{x}, \dot{y}]^T$. The system is governed by the state transition equation that is the result of discretizations of forward Euler:

$$\mathbf{x_{k+1}} = \mathbf{A}\mathbf{x_k} + \mathbf{B}\mathbf{u} + \mathbf{w_k}. \tag{26}$$

Based on the constant velocity model, the state transition matrix $\mathbf{A}$ and the control input matrix $\mathbf{B}$ are defined by the time step Δt:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} \frac{\Delta t^2}{2} & 0 \\ 0 & \frac{\Delta t^2}{2} \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}. \tag{27}$$

The control vector is given by the acceleration $\mathbf{u} = [\ddot{x}, \ddot{y}]^\top$ and $\mathbf{w_k} \sim \mathcal{N}(0, \mathbf{Q})$ represents the process perturbation vector with covariance matrix $\mathbf{Q}$. The process noise covariance matrix was empirically defined as $\mathbf{Q} = \text{diag}(0.05^2, 0.05^2, 0.01^2, 0.001^2)$ to account for unmodeled physical disturbances during locomotion, such as mechanical backlash in the servomotors. The

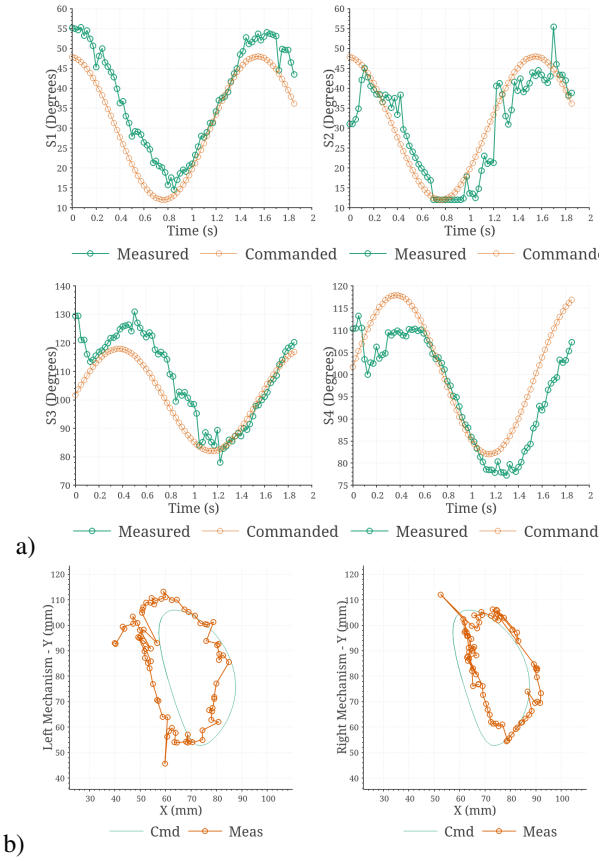

Fig. 6. Cycloidal gait. (a) Angular trajectories of actuators over time. (b) Task space, comparing commanded cycloidal paths versus measured trajectory

measurement vector $\mathbf{z_k} \in \mathbb{R}^2$, containing the noisy positional sensor readings, is modeled as:

$$\mathbf{z_k} = \mathbf{H}\mathbf{x_k} + \mathbf{v_k} \tag{28}$$

Where the observation matrix $\mathbf{H}$ maps the true state space into the observed space, tracking only the x and y coordinates:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \tag{29}$$

The measurement noise $\mathbf{v_k}$ is bounded by the sensor's covariance matrix $\mathbf{R}$. Based on the system's experimental characterization, the noise standard deviation for the measurements was established at $\sigma = 0.01$, defining the observation covariance as $\mathbf{R} = \text{diag}(0.01^2, 0.01^2)$. The Kalman filter operates recursively. First, the prediction step projects the current state and error covariance matrix $\mathbf{P_k}$ forward:

$$\hat{\mathbf{x}}_{k+1} = \mathbf{A}\hat{\mathbf{x}}_k + \mathbf{B}\mathbf{u}, \tag{30}$$

$$\mathbf{P}_{k+1} = \mathbf{A}\mathbf{P}_k\mathbf{A}^T + \mathbf{Q}. \tag{31}$$

Thus, the update step corrects the estimate using a new measurement $\mathbf{z_{k+1}}$ by calculating the Kalman Gain $\mathbf{K_{k+1}}$:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}\mathbf{H}^T(\mathbf{H}\mathbf{P}_{k+1}\mathbf{H}^T + \mathbf{R})^{-1} \tag{32}$$

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1} + \mathbf{K}_{k+1}(\mathbf{z}_{k+1} - \mathbf{H}\hat{\mathbf{x}}_{k+1}) \quad (33)$$

$$\mathbf{P}_{k+1} = (\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H})\mathbf{P}_{k+1} \quad (34)$$

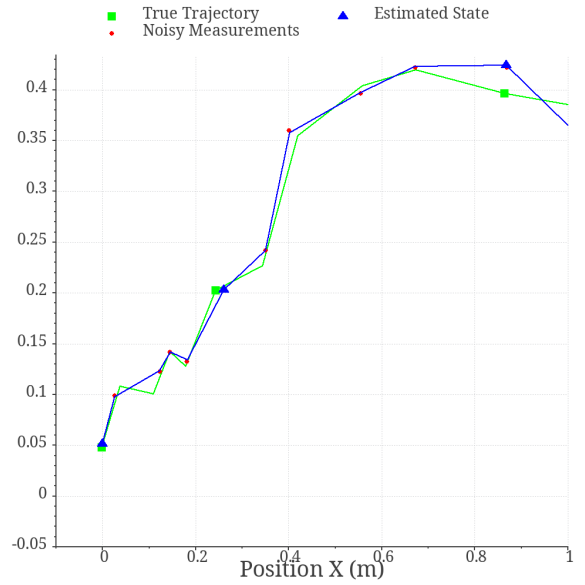

Fig. 7. Motion state estimation using a discrete-time Kalman filter.

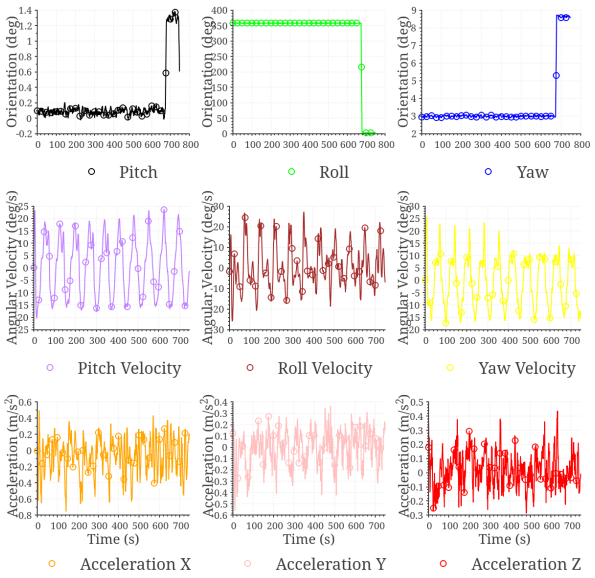

Fig. 8. Raw IMU data from walking experiment.

VII. Conclusion

In this study, we successfully developed a rigorous analytical framework for the kinematics of an experimental five-bar bipedal prototype. By establishing a deterministic, closed-form mapping between joint and task spaces, we eliminated the computational overhead typically associated with iterative numerical solvers. The alignment between our simulations and experimental results although it is still undergoing, but it is found that the proposed model provides a reliable foundation for real-time trajectory tracking and precise foot placement in closed-loop robotic systems. Future work involves the inclusion of dynamic control and experimental multi-gait walking. Furthermore, an upgrade towards the Unscented Kalman filter to combine IMU data and the limbs angular transducers.

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